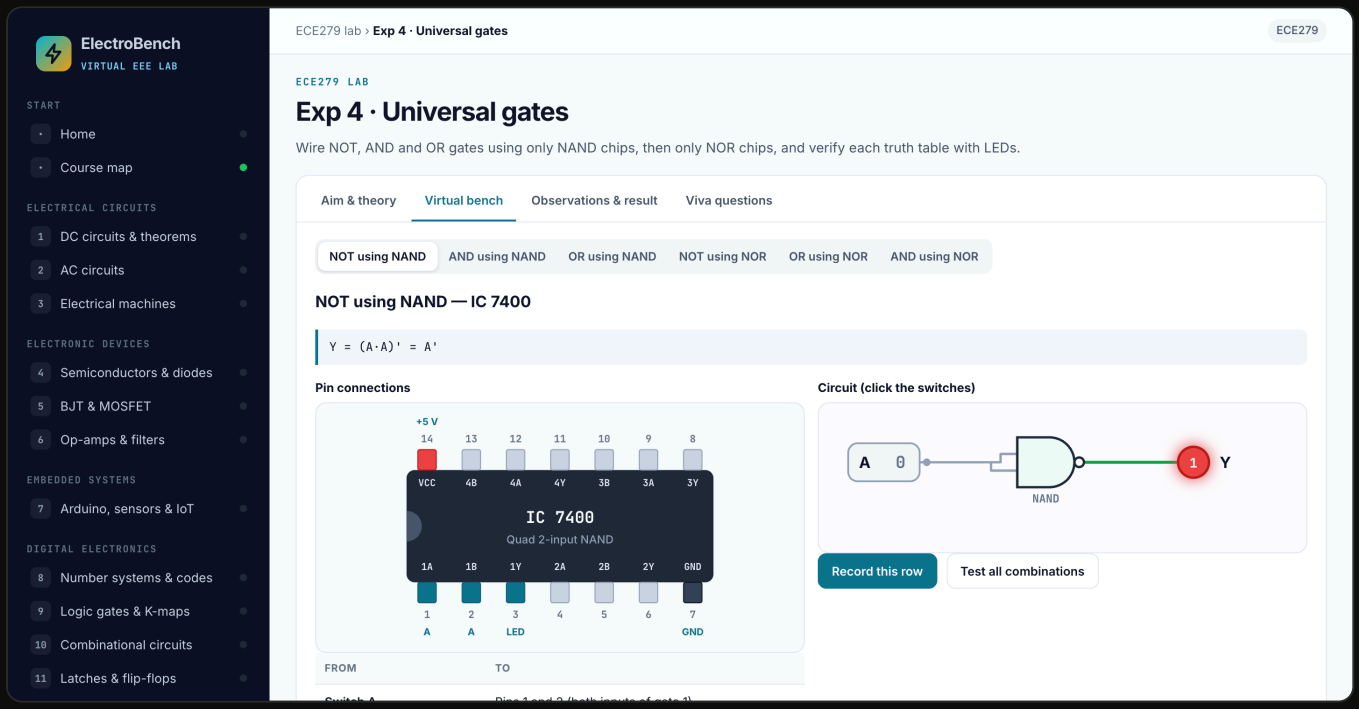
ElectroBench

An interactive workbench for basic electrical and electronics engineering. Every unit of ECE131 is a working model you can change, and every lab experiment has a virtual bench whose readings fill in a printable lab record.



STUDENT	Lukshya Singh Supyal	REGISTRATION NO.	
PROGRAMME	B.Tech CSE, Semester 5	COURSE	ECE131 Basic Electrical & Electronics Engineering (2026-27). Also covers the ECE279 lab and the earlier ECE249 syllabus
LIVE WEBSITE	lushadowx.github.io/ElectroBench	SOURCE CODE	github.com/LuShadowX/ElectroBench

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08 **ECE131 Units V and VI**
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10 **Earlier syllabus (ECE249)**
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How the important parts work

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GitHub Pages, what I learned

> Every screenshot in this report was taken from ElectroBench while using it, with real readings taken on the virtual benches. Every code extract is copied from the source files, with file names and line numbers.

Course and scope

The project is built on the ECE131 Basic Electrical and Electronics Engineering syllabus for session 2026-27: all six units and all six course outcomes are covered, and chapters 04 to 08 follow the syllabus unit by unit. ECE131 is a theory course (L:3 T:1 P:0), so the practical side is covered by the seven experiments of the ECE279 lab, each with a virtual bench. The same course was earlier offered as ECE249, which had more digital electronics (number systems, combinational and sequential circuits, counters); those topics are built too and shown in chapter 10, so the project matches either version of the course.

01 · PROJECT OVERVIEW

Learning electronics by changing it

In a textbook a circuit is a fixed picture with one set of values. ElectroBench turns every circuit into something you can change: move a slider and the mesh currents, the diode curve, the filter's cut-off or the flip-flop's output react immediately. The lab experiments get a virtual bench with a power supply, multimeter, CRO and TTL chips, so the procedure can be practised before (or after) the real lab.

12
topic pages

7
lab experiments with virtual benches

87
interactive screens (tabs)

~5,300
lines of JavaScript, HTML and CSS

Sections

SECTION	PAGES	COURSES
Electrical circuits	DC circuits and theorems · AC circuits · Electrical machines	ECE131 Units I–III
Electronic devices	Semiconductors and diodes · BJT and MOSFET · Op-amps and filters	ECE131 Units IV and V
Embedded systems	Arduino, sensors and IoT	ECE131 Unit VI
Digital electronics	Logic gates · Number systems · Combinational circuits · Latches and flip-flops · Registers and counters	ECE131 Unit IV (gates); earlier ECE249 Units 3–6
ECE279 lab	Seven experiments, each with aim and theory, virtual bench, observations and result, viva questions, plus a printable lab record	ECE279
Course map	Every syllabus topic of the three courses with a link to the exact page and tab	All

What makes it different from a set of notes

- **The models are real calculations.** The DC page solves the network with nodal analysis, the diode bench solves the Shockley equation for the operating point, and the rectifier page simulates the capacitor charging and discharging in time.
- **Digital circuits are simulated at gate level with gate delays.** Every adder, multiplexer and flip-flop is a netlist of gates. Because each gate has a delay, a JK flip-flop with the clock held high really oscillates: the race-around condition is something you see, not just read.
- **Experiments produce a lab record.** Readings taken on the bench are saved and fill in the observation tables, the result is written from those readings, and the Lab record page prints all seven experiments as one file.

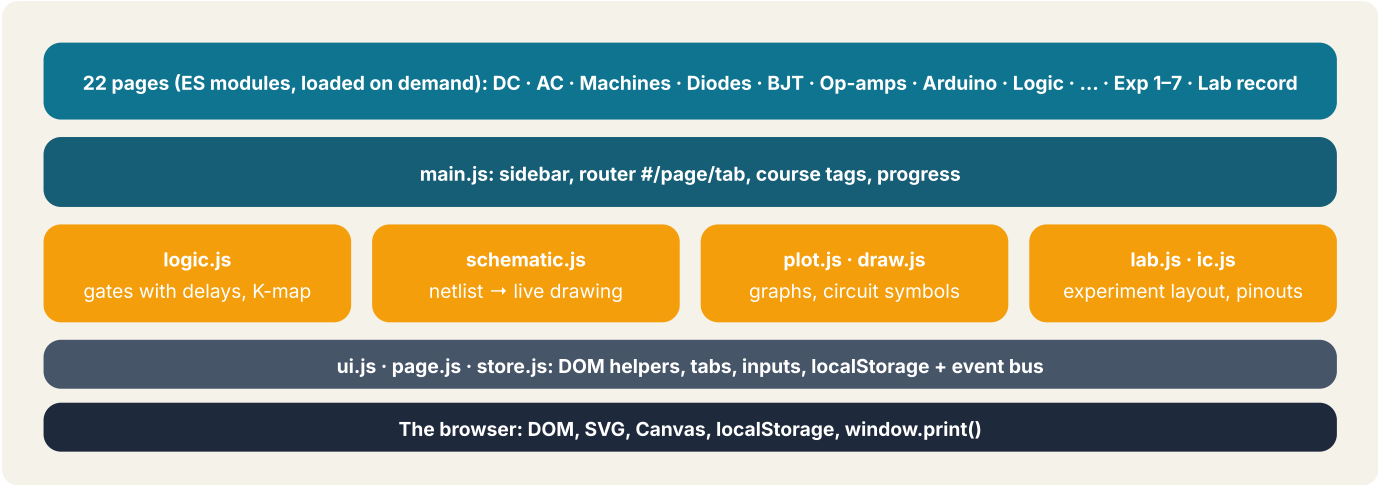
Technology

Plain HTML, CSS and JavaScript (ES modules). There is no framework, no build step and no server. Drawings are SVG generated in code, the CRO screen is a canvas, and progress and readings are stored in the browser's local storage. The site is hosted on GitHub Pages.

02 · HOW IT'S BUILT

Architecture

Each page is an ES module that is loaded only when it is opened. Pages share a small set of engines: a logic simulator, a schematic drawer, a plotter, a circuit-symbol kit and the experiment layout.



Files

FILE	JOB	LINES
`js/main.js`	Sections, sidebar with progress ticks, router, course tags on every page	135
`js/core/logic.js`	Gate-level simulator with unit delays, Boolean parser, Quine–McCluskey minimisation, SOP circuit builder	270
`js/core/schematic.js`	Lays out any netlist by logic depth and draws it, colouring wires that carry 1	204
`js/core/plot.js`	SVG graphs with linear or log axes, reference lines, markers and a hover readout	108
`js/core/draw.js`	Circuit symbols: resistor, sources, diode, LED, Zener, capacitor, inductor, op-amp	136
`js/core/lab.js`	The four-tab experiment layout and saved observations	71
`js/core/ic.js`	Pinouts of the 7400, 7402, 7408, 7410, 7432 and 7486	35
`js/pages/dc.js`	Ohm, Kirchhoff, dividers, series–parallel reducer, star–delta, mesh and nodal, theorems	308
`js/pages/semis.js`	Doping, PN junction, V-I, Zener, varactor, rectifier simulation, testing	281
`js/pages/sequential.js`	Gate-level latches, flip-flops with timing diagrams, conversion solver	234
`js/pages/lab1.js` … `lab7.js`	The seven experiments: bench, observation tables, result, viva	—
`css/style.css`, `css/pages.css`	Layout, components, circuit and instrument styles, print styles for the lab record	—

One idea, many pages

Because every digital circuit is just a netlist (a list of gates and what feeds each input), the same simulator and the same schematic drawer power the gate page, the universal-gate page, all combinational circuits, the K-map solver's generated circuit, the latches and Experiments 4, 5 and 6. The circuit drawn on the screen is the circuit that is simulated, so they cannot disagree.

03 · HOME AND COURSE MAP

Finding your way

The sidebar groups pages the way the syllabus does and ticks each page once it has been opened. Every page shows which course units it covers as tags in the top bar, for example "ECE131 U-IV · ECE249 U-1 · Lab 3" on the semiconductor page, so a page can be traced back to the syllabus from either side.

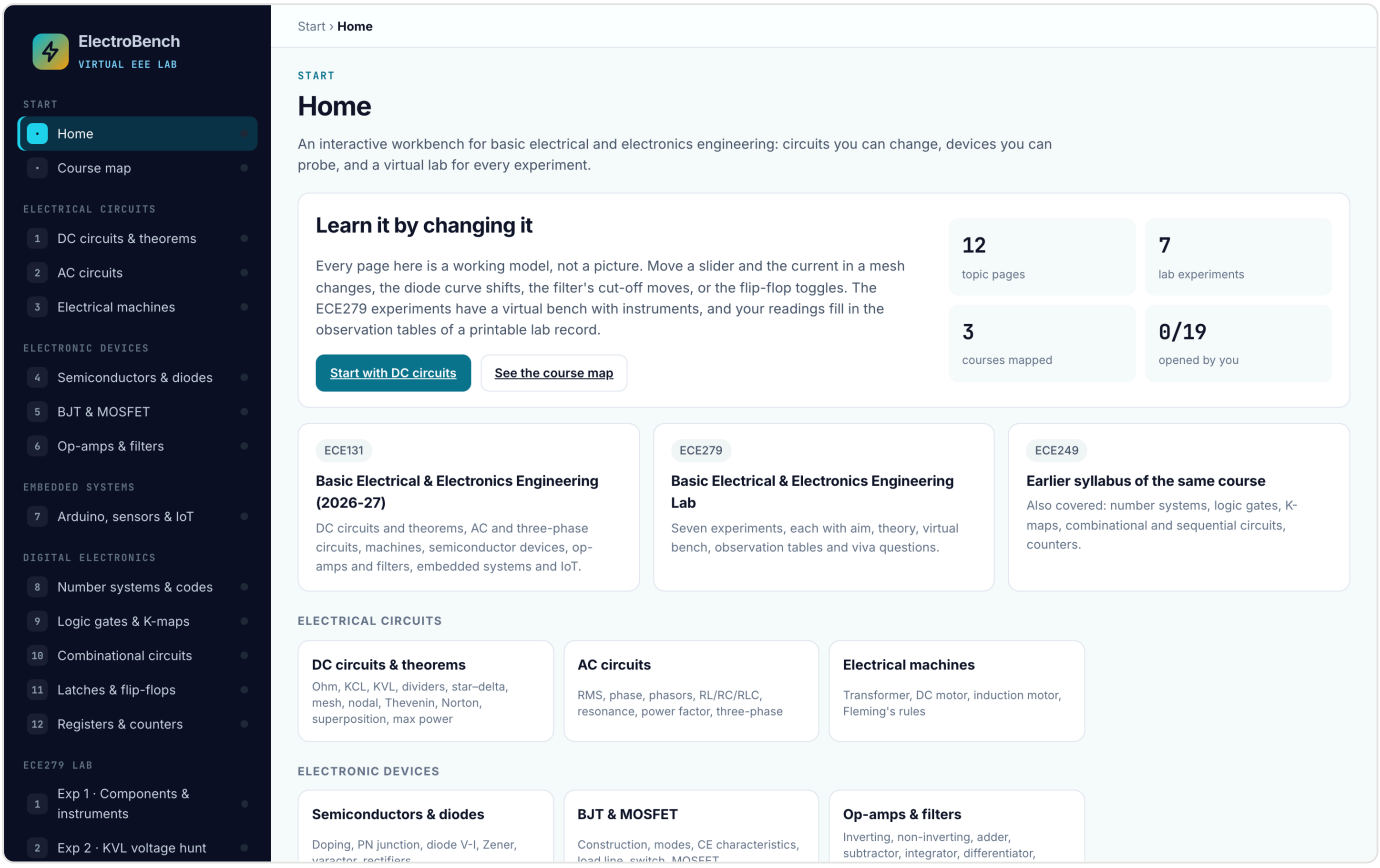
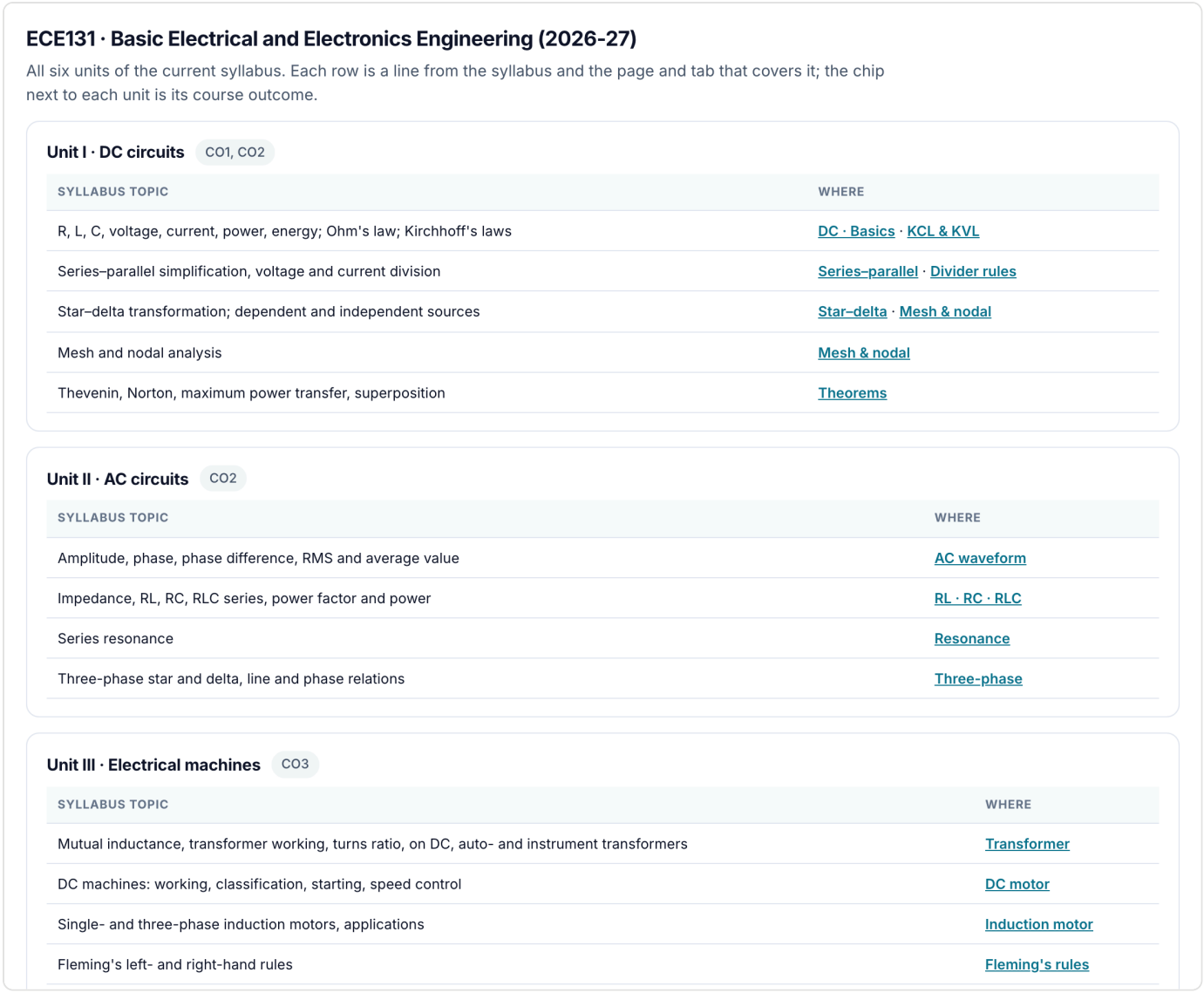


Fig. 1 · **Home.** Sections, the three courses, progress, and a "continue" button that opens the first page not yet visited.



ECE131 · Basic Electrical and Electronics Engineering (2026-27)

All six units of the current syllabus. Each row is a line from the syllabus and the page and tab that covers it; the chip next to each unit is its course outcome.

Unit I · DC circuits CO1, CO2

SYLLABUS TOPIC	WHERE
R, L, C, voltage, current, power, energy; Ohm's law; Kirchhoff's laws	DC · Basics · KCL & KVL
Series–parallel simplification, voltage and current division	Series–parallel · Divider rules
Star–delta transformation; dependent and independent sources	Star–delta · Mesh & nodal
Mesh and nodal analysis	Mesh & nodal
Thevenin, Norton, maximum power transfer, superposition	Theorems

Unit II · AC circuits CO2

SYLLABUS TOPIC	WHERE
Amplitude, phase, phase difference, RMS and average value	AC waveform
Impedance, RL, RC, RLC series, power factor and power	RL · RC · RLC
Series resonance	Resonance
Three-phase star and delta, line and phase relations	Three-phase

Unit III · Electrical machines CO3

SYLLABUS TOPIC	WHERE
Mutual inductance, transformer working, turns ratio, on DC, auto- and instrument transformers	Transformer
DC machines: working, classification, starting, speed control	DC motor
Single- and three-phase induction motors, applications	Induction motor
Fleming's left- and right-hand rules	Fleming's rules

Fig. 2 · **Course map, ECE131 tab.** Every line of the syllabus, unit by unit with its course outcome, links straight to the page and tab that covers it. The lab and the earlier ECE249 syllabus have their own tabs. The full mapping is in chapter 11.

Fundamentals of DC circuits

Unit I: resistance, inductance, capacitance, voltage, current, power and energy, Ohm's law, Kirchhoff's laws, series and parallel simplification, voltage and current division, star–delta transformation, dependent and independent sources, mesh and nodal analysis, Thevenin, Norton, maximum power transfer and superposition. CO1 CO2

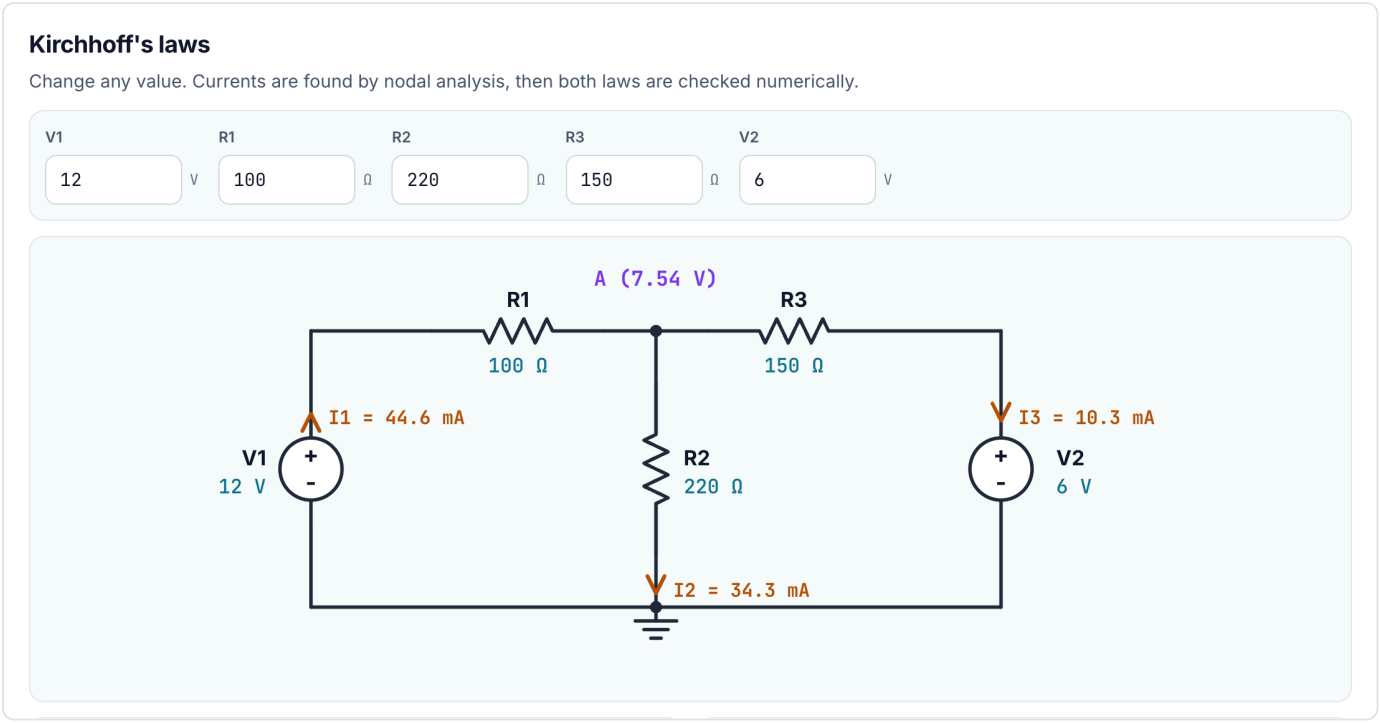


Fig. 3 · **KCL and KVL.** A two-source network solved by nodal analysis. Change any source or resistor and the currents into node A and the voltages around each loop are added up live, so you can see both laws hold for every value.

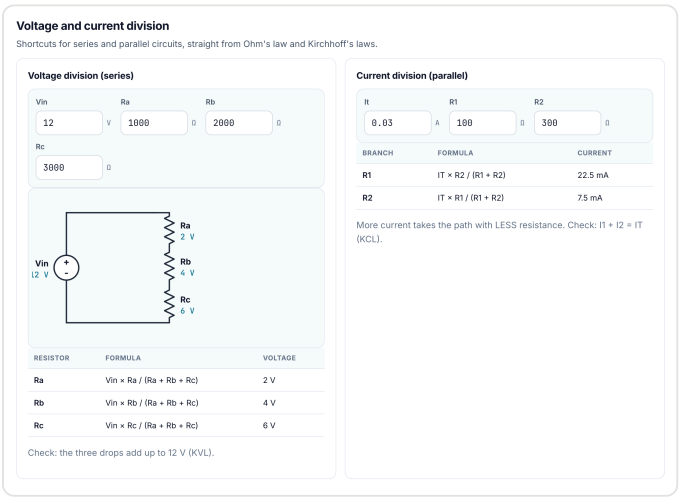


Fig. 4 · **Divider rules.** Voltage and current division with the formula filled in with the current values.

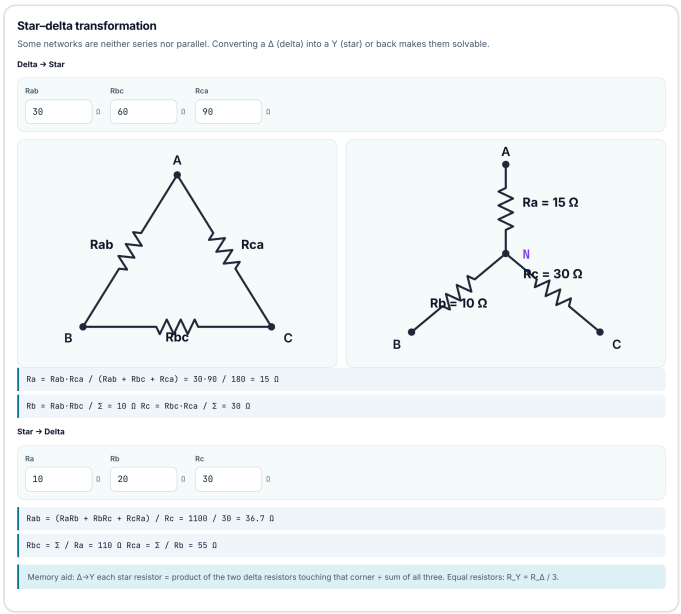


Fig. 5 · **Star–delta.** Convert in either direction with the formulas filled in.

Mesh and nodal analysis

Two systematic methods that work for any circuit. Mesh uses KVL and loop currents. Nodal uses KCL and node voltages. Same circuit, same answer.

V1	R1	R2	R3	V2
12 V	100 Ω	220 Ω	150 Ω	6 V

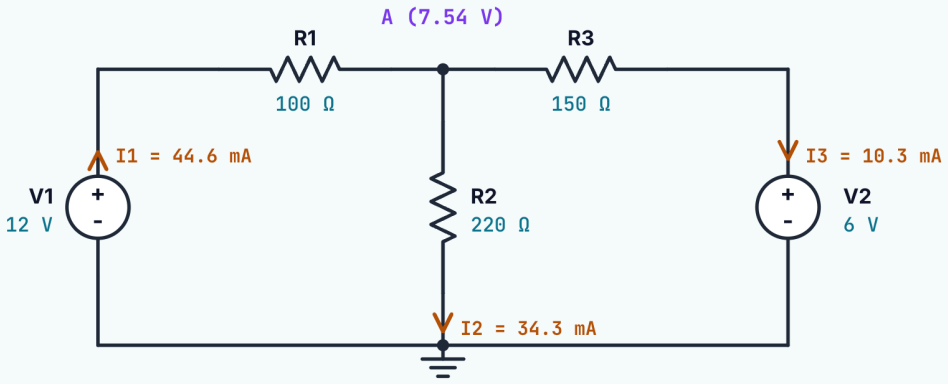
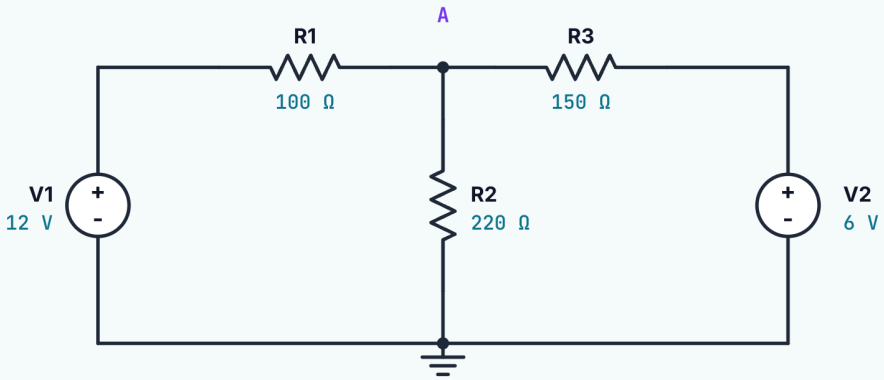


Fig. 6 · **Mesh and nodal analysis.** The equations written out and solved, plus dependent and independent sources.

Network theorems

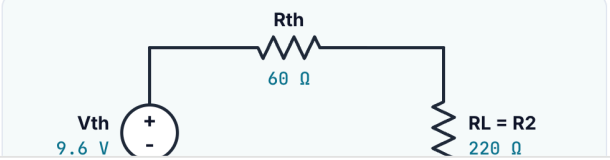
Thevenin, Norton, superposition and maximum power transfer, all on the same two-source circuit so you can see they agree.

V1	R1	R2	R3	V2
12 V	100 Ω	220 Ω	150 Ω	6 V



Thevenin equivalent (seen by R2)

1. Remove the load R2 and find the open-circuit voltage at A: `Vth = (V1·R3 + V2·R1)/(R1 + R3) = 9.6 V`
2. Replace sources by short circuits and find the resistance at A: `Rth = R1 ∥ R3 = 60 Ω`
3. Reconnect the load: `IL = Vth/(Rth + R2) = 34.3 mA`



Norton equivalent

The same circuit as a current source in parallel with a resistor.

```
IN = Vth / Rth = 160 mA
```

```
RN = Rth = 60 Ω
```

```
IL = IN · RN/(RN + RL) = 34.3 mA
```

Matches the full solution: I(R2) = 34.3 mA ✓

Fig. 7 · **Theorems.** Thevenin, Norton, superposition and maximum power transfer on the same network, so the answers can be compared. The power curve peaks where RL = Rth.

Fundamentals of AC circuits

Unit II: alternating voltage and current, amplitude, phase, phase difference, RMS and average value, complex impedance, RL, RC and RLC series circuits, resonance, power factor and power, and three-phase star and delta connections. CO2



Fig. 8 · AC waveform. Amplitude, phase difference, RMS and average values.

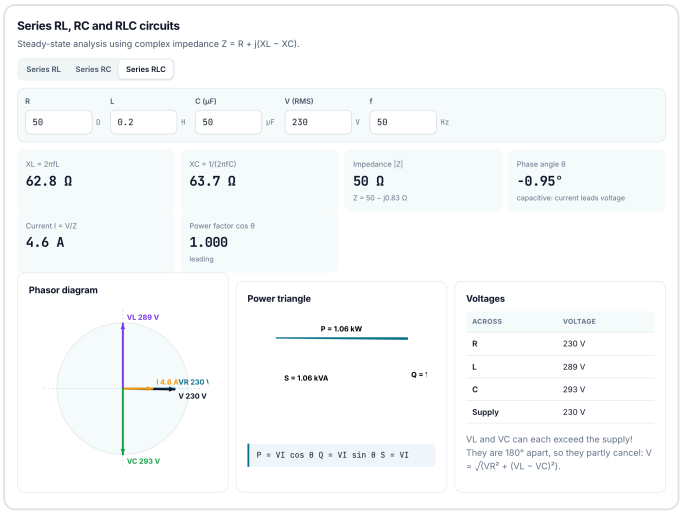


Fig. 9 · RL, RC and RLC. Impedance, phasor diagram, power factor and real, reactive and apparent power.

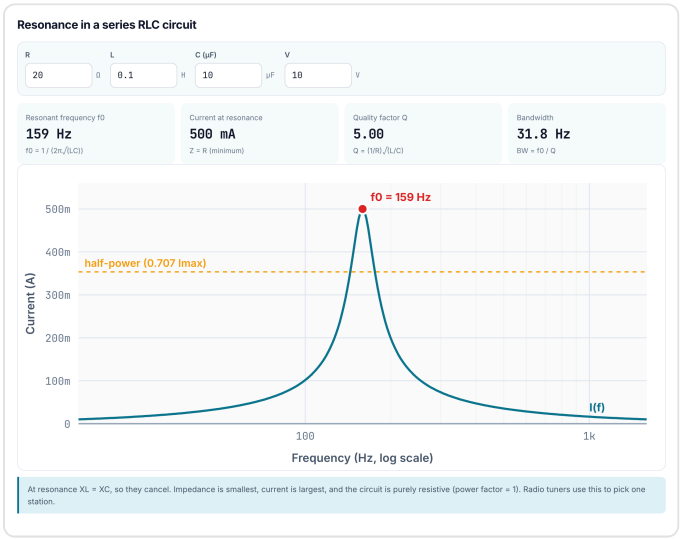


Fig. 10 · Resonance. Current against frequency; f0, bandwidth and Q.

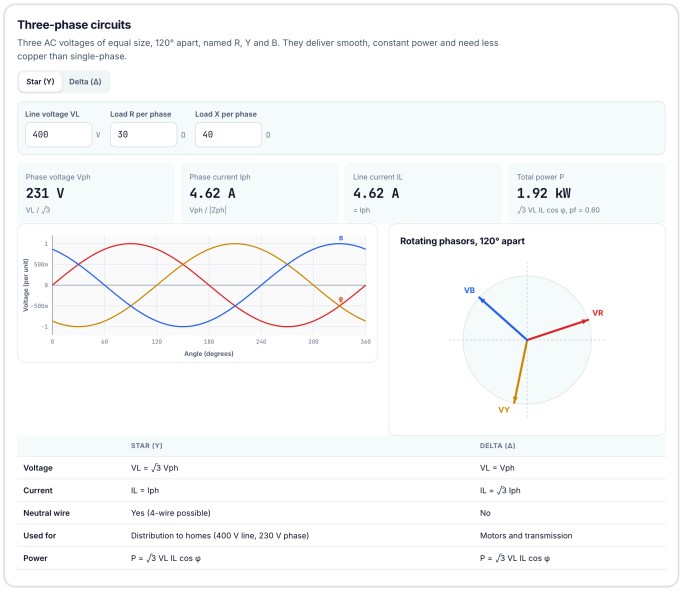


Fig. 11 · Three-phase. Star and delta, line and phase relations.

Fundamentals of electrical machines

Unit III: mutual inductance and the transformer (working, turns ratio, transformer on DC, auto- and instrument transformers), DC machines (principle, classification, starting, speed control), single- and three-phase induction motors and Fleming's rules. CO3

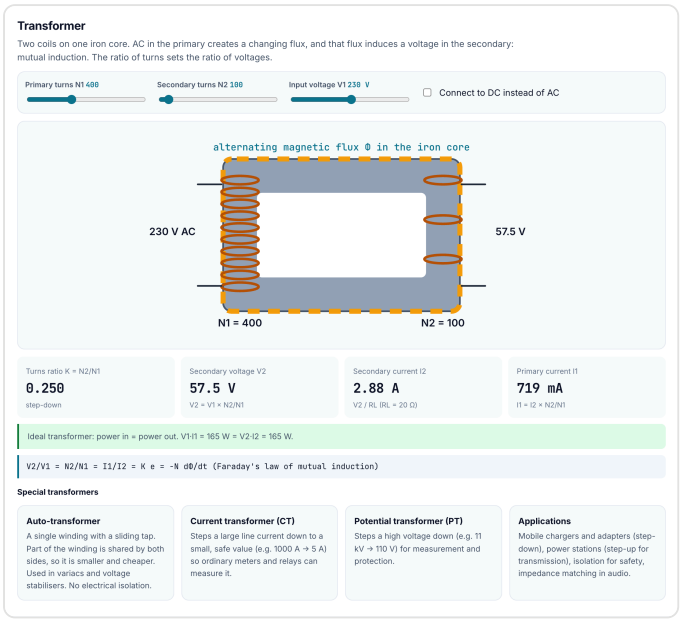


Fig. 12 · Transformer. Mutual induction with animated flux, turns ratio, and what happens on DC; auto- and instrument transformers.

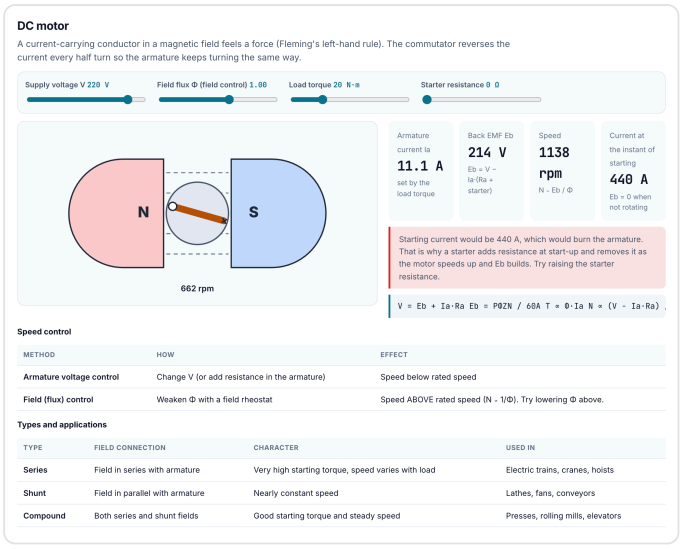


Fig. 13 · DC motor. Working principle, types, starting and speed control (N ∝ (V − IaRa)/Φ).

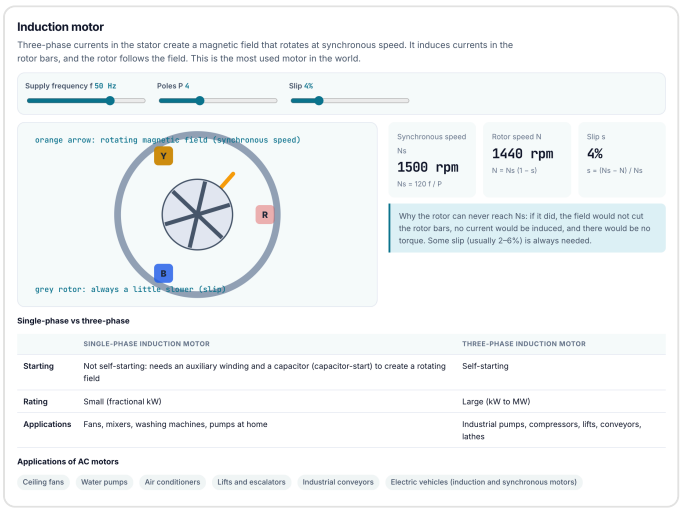


Fig. 14 · Induction motor. The rotating field of three phases, slip, single- and three-phase motors.

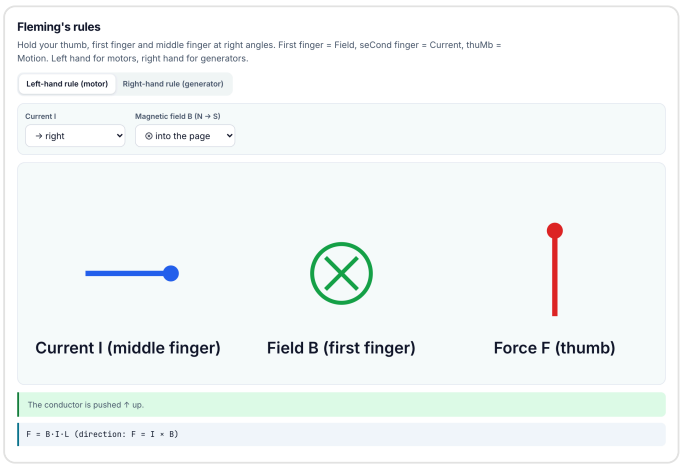


Fig. 15 · Fleming's rules. Set the field and current directions; the force comes from the cross product.

Fundamentals of semiconductor devices

Unit IV: introduction to semiconductors, the PN junction diode and its characteristics, the Zener and varactor diodes, testing of diode and BJT, half wave, full wave and bridge rectifiers, MOSFET representation and characteristics, logic gates, and applications of diodes, transistors and MOSFETs. CO4

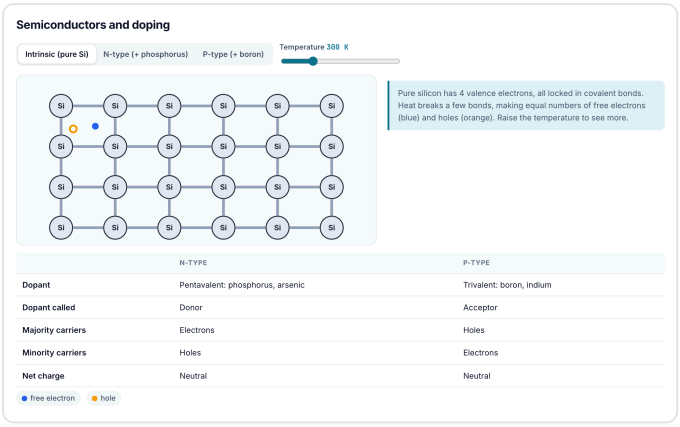


Fig. 16 · **Intrinsic and extrinsic.** A silicon lattice. Add phosphorus (N-type) or boron (P-type) and the free electrons or holes appear.

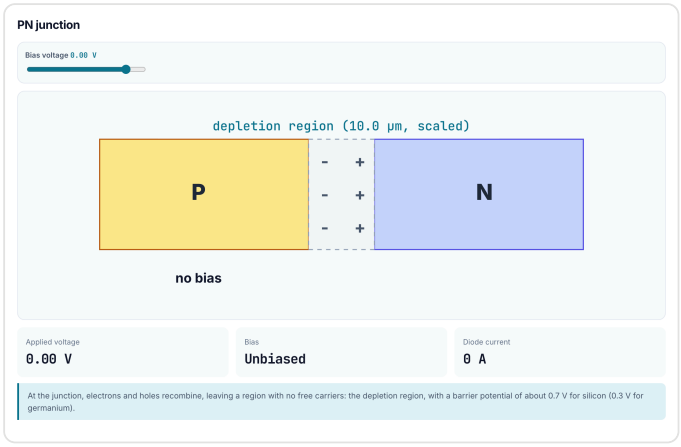


Fig. 17 · **PN junction.** The depletion region narrows in forward bias and widens in reverse bias.

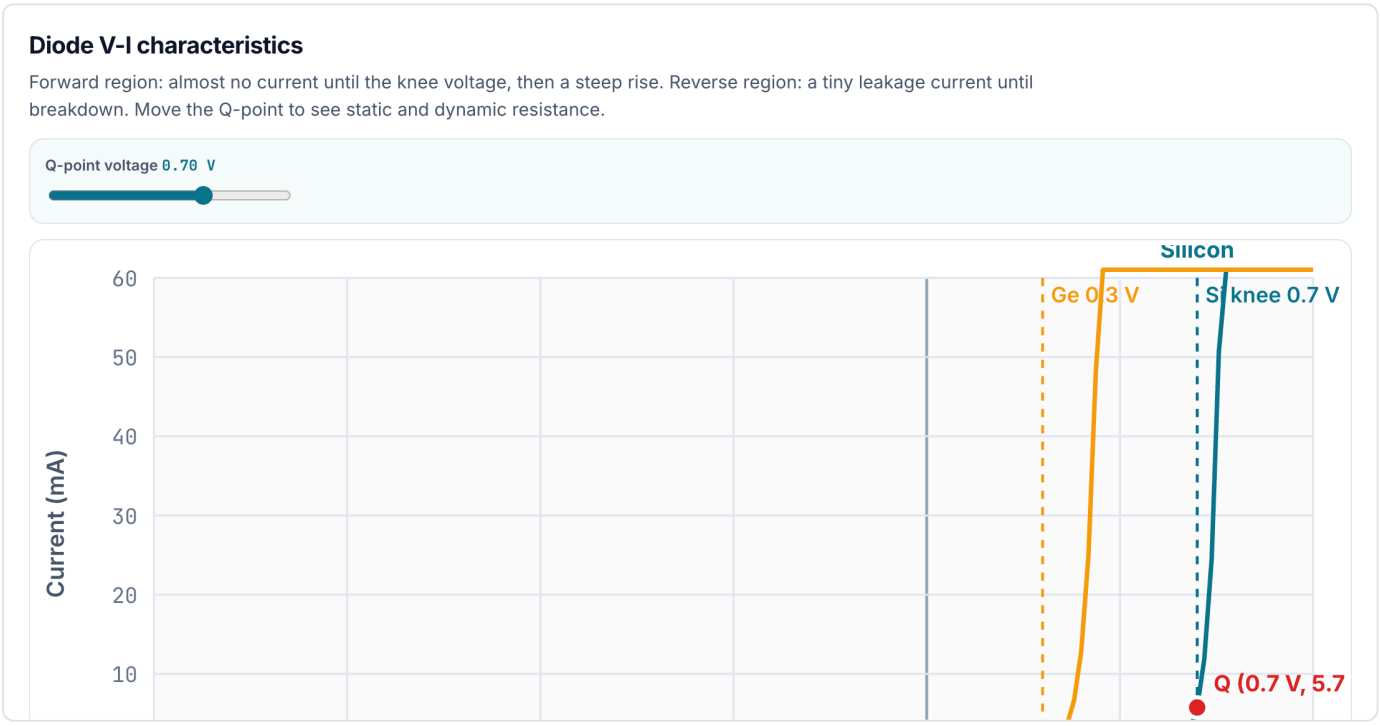


Fig. 18 · **V-I characteristic.** Calculated from the Shockley equation; silicon and germanium knees compared.

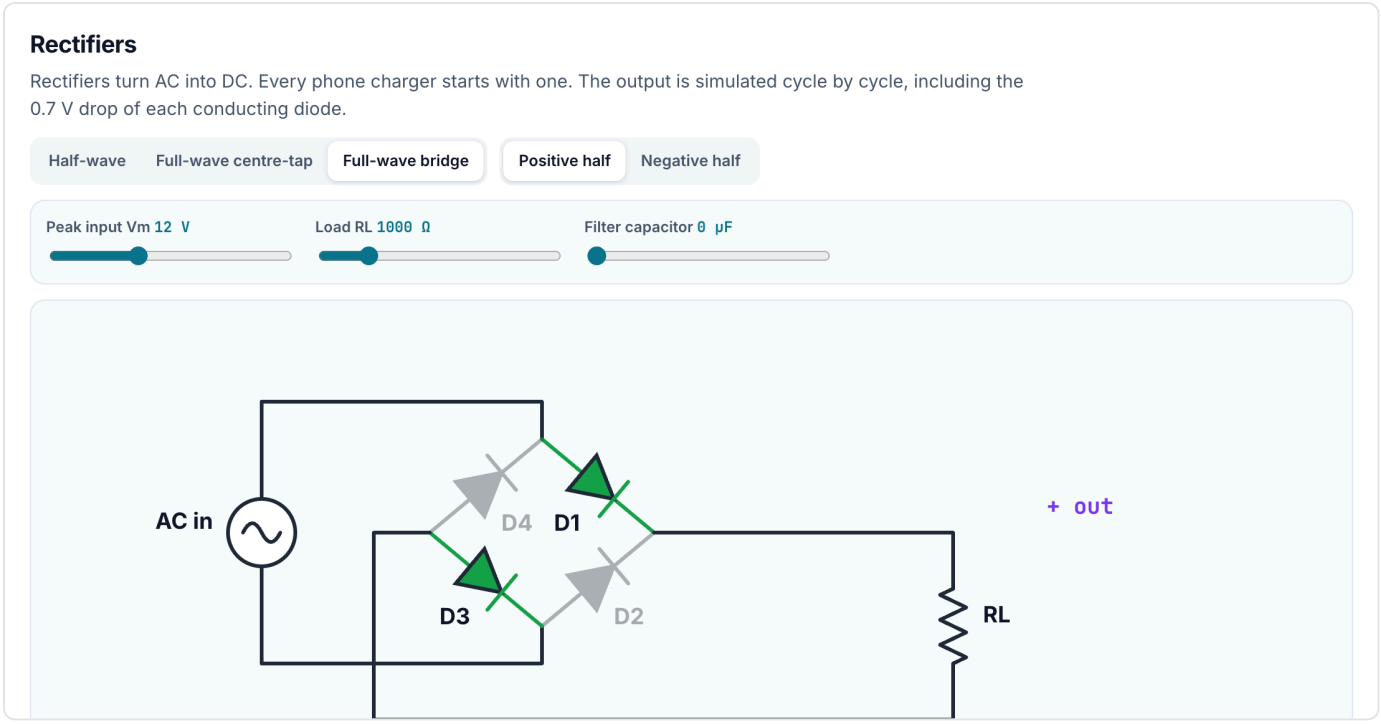


Fig. 19 · **Rectifiers.** Half wave, centre-tap and bridge, simulated in time. Add a filter capacitor and the ripple and DC level are measured from the simulated waveform.

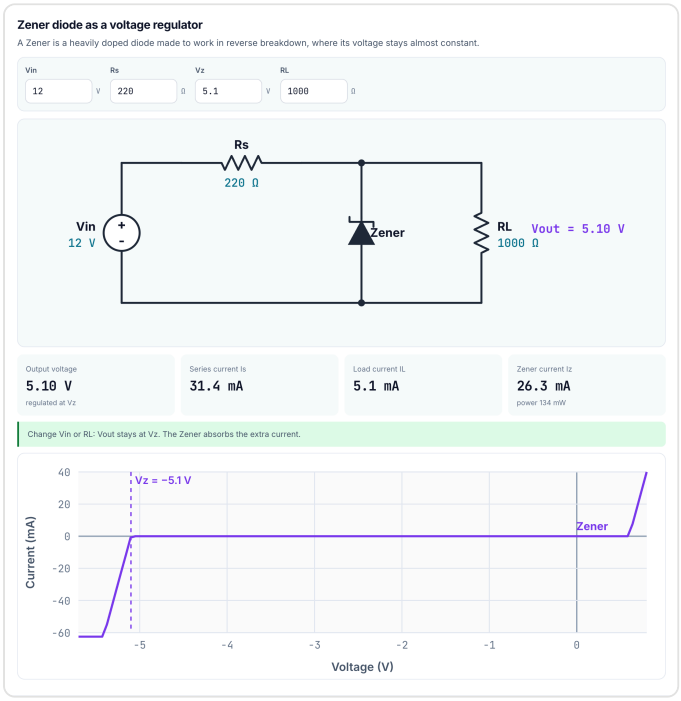


Fig. 20 · **Zener.** Regulator with line and load changes.

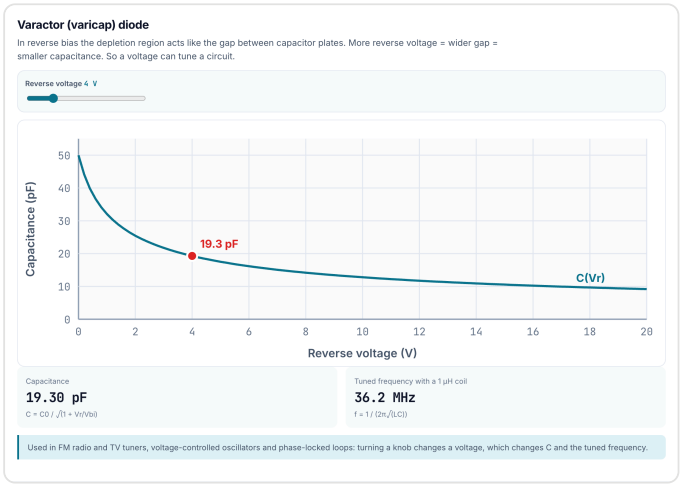


Fig. 21 · **Varactor.** Capacitance falls with reverse voltage.

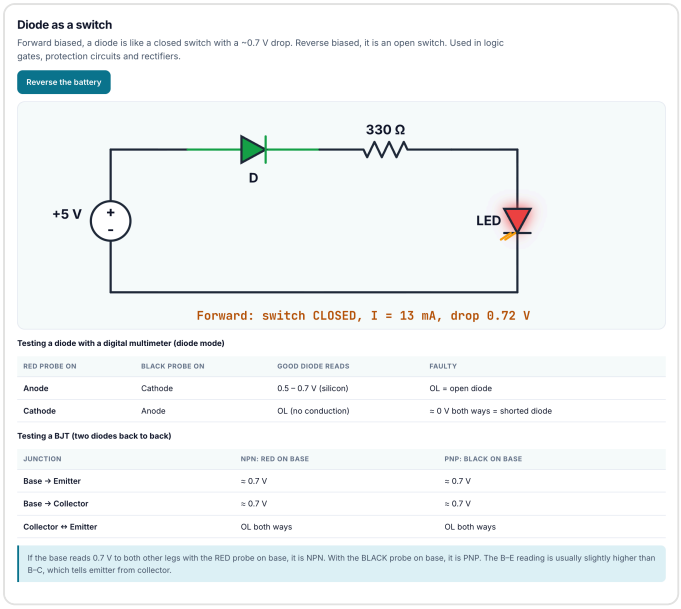


Fig. 22 · **Diode as a switch, and testing.** ON and OFF states, and how to test a diode and a BJT with a multimeter.



Fig. 23 · **BJT construction and modes.** NPN and PNP, and the four operating modes from the two junction biases.

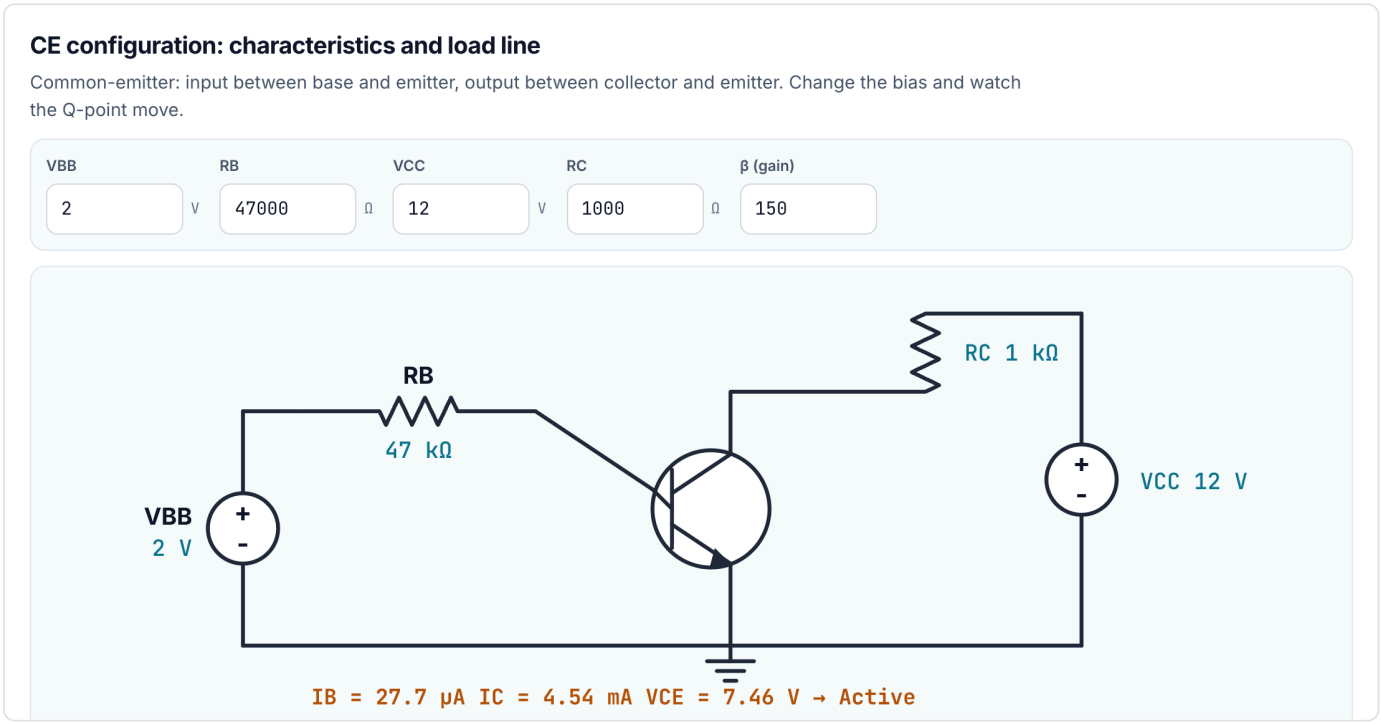


Fig. 24 · **CE characteristics.** Output curves for several base currents (with the Early effect), a load line from Vcc and Rc, and the Q-point moving as you change the base current.



Fig. 25 · **MOSFET.** Symbols and square-law characteristics.

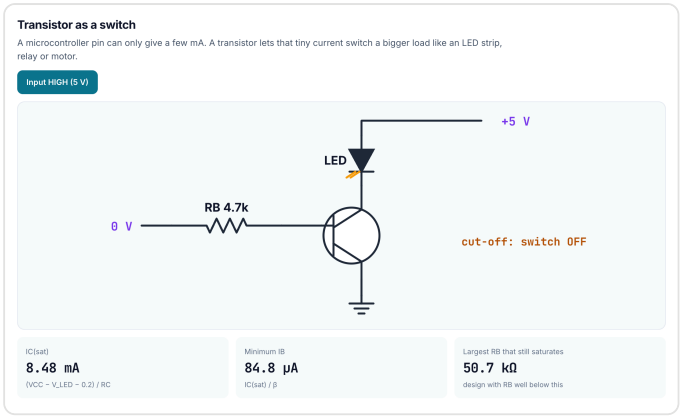


Fig. 26 · **BJT as a switch**, with the base resistor calculation for saturation.

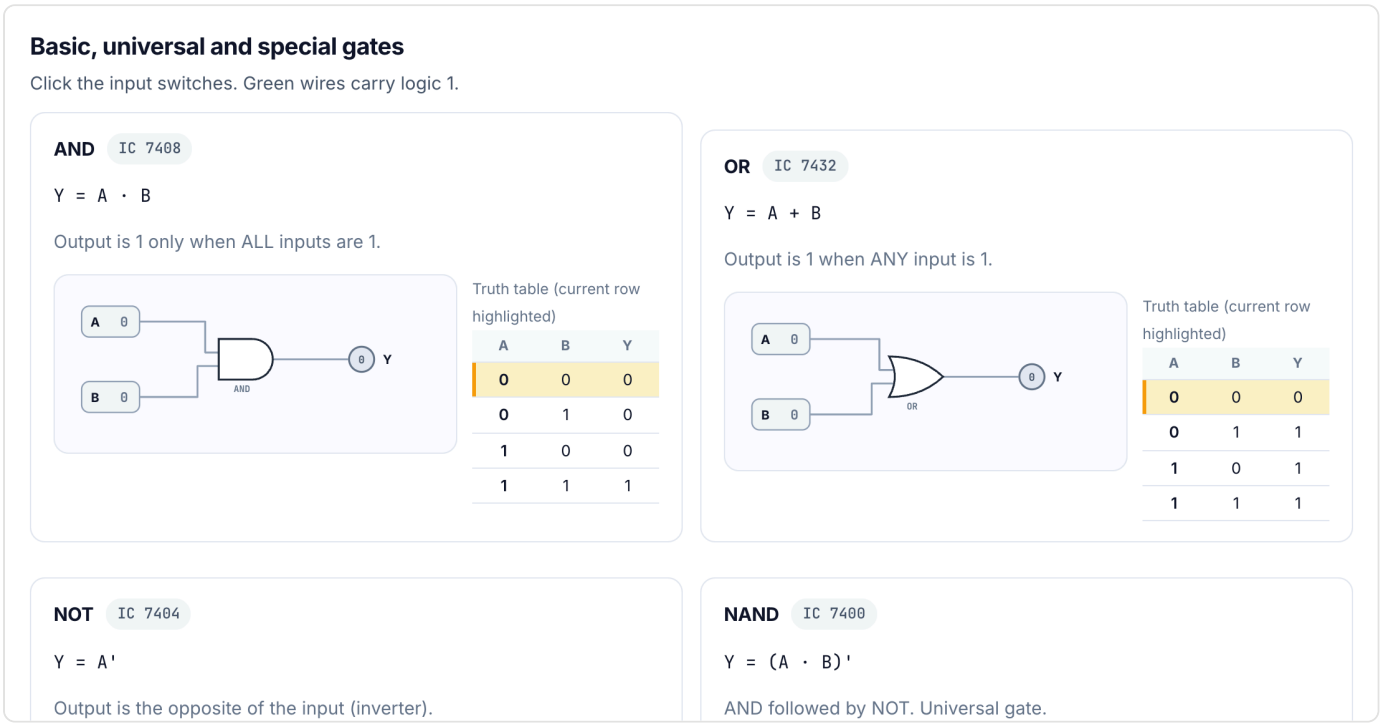


Fig. 27 · **Logic gates.** All seven gates as live circuits with their IC numbers and truth tables.

Op-amps, filters and embedded systems

Unit V: filters, op-amp properties and the virtual ground, inverting and non-inverting amplifiers, adder and subtractor, integrator and differentiator, active filters and the comparator in anti-lock braking. Unit VI: embedded systems, microprocessor vs microcontroller, Arduino and sensors, and IoT and cloud in industry, healthcare, agriculture, automobiles and the smart grid. CO5 CO6

Unit V · Filters and operational amplifiers

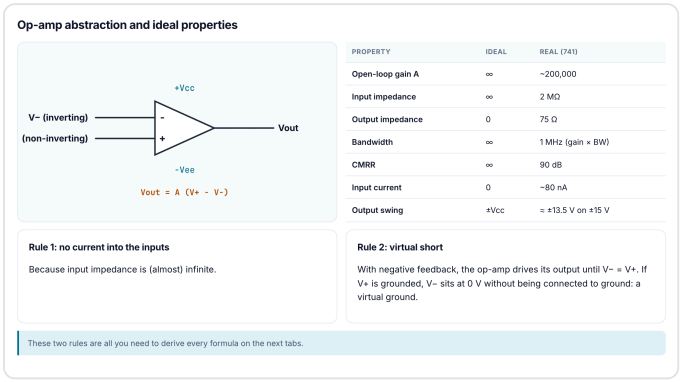


Fig. 28 · **Op-amp properties.** Ideal vs 741, virtual ground.

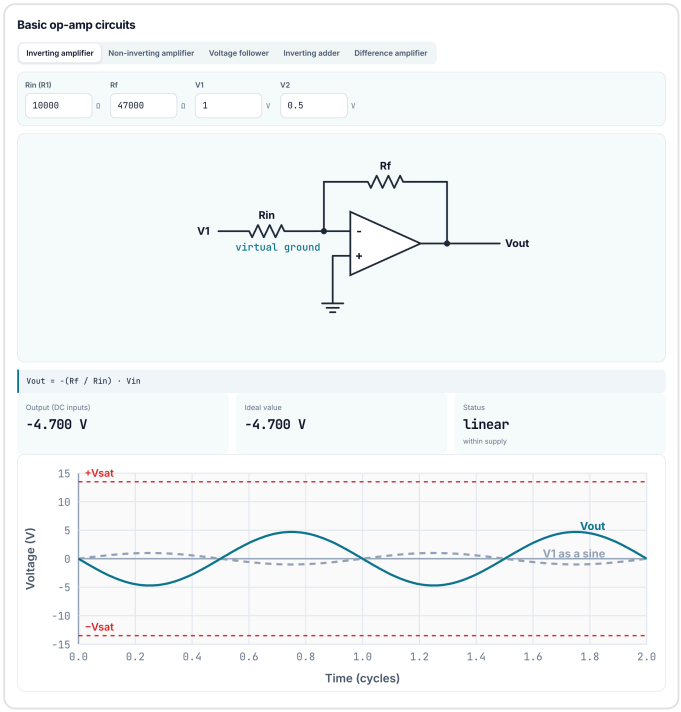


Fig. 29 · **Configurations.** Inverting, non-inverting, follower, adder and subtractor, with saturation at ±13.5 V.

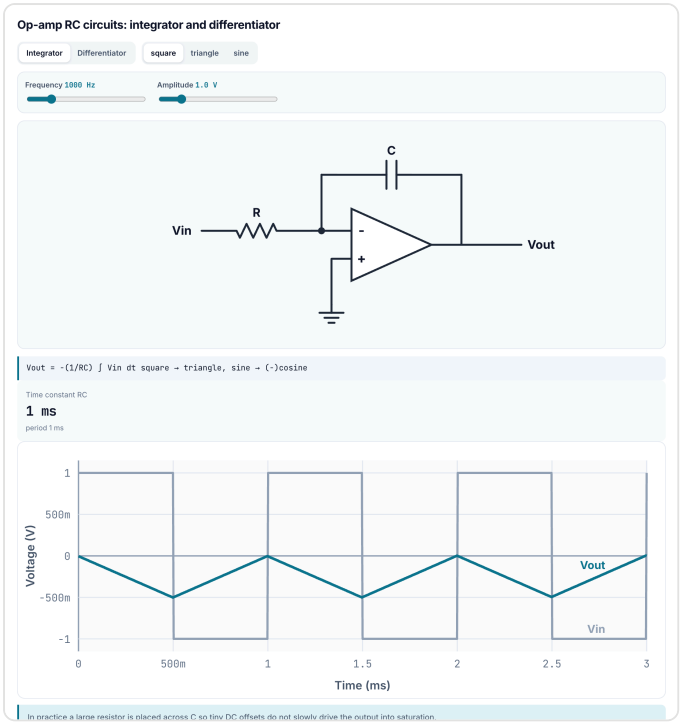


Fig. 30 · **Integrator and differentiator.** Square in, triangle out (and the reverse).



Fig. 31 · **Filters.** Low-, high-, band-pass and band-stop on a log frequency axis, passive or active.

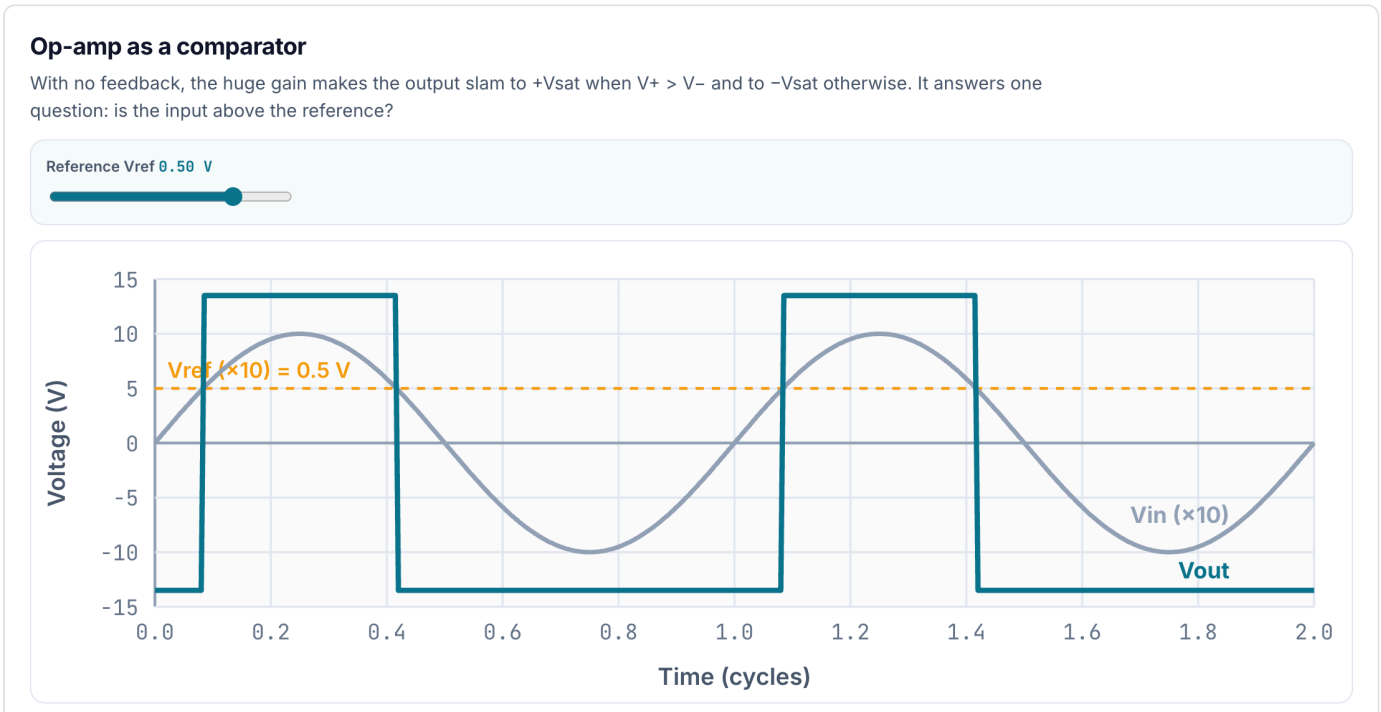


Fig. 32 · **Comparator and ABS.** The comparator switches when the input crosses Vref. Below, an anti-lock braking simulation: the comparator watches wheel slip and releases the brake briefly whenever the wheel starts to lock.

Unit VI · Embedded systems and IoT

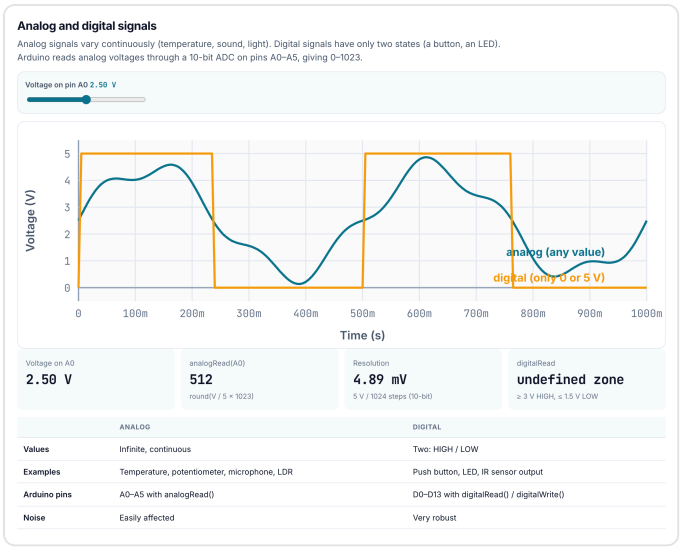


Fig. 33 · **Analog vs digital.** A sine wave sampled and quantised by a 10-bit ADC, as analogRead does.

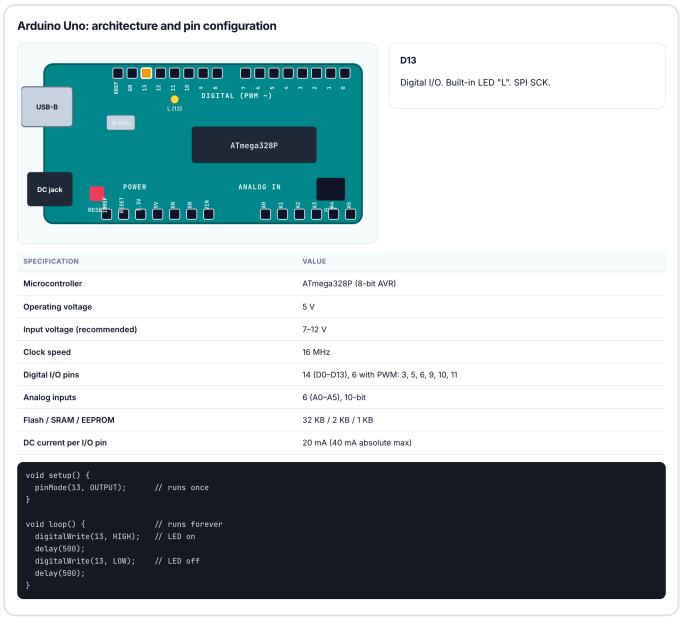


Fig. 34 · **Arduino Uno.** Click any pin on the drawn board to see what it does (digital, PWM, analog, power, serial).

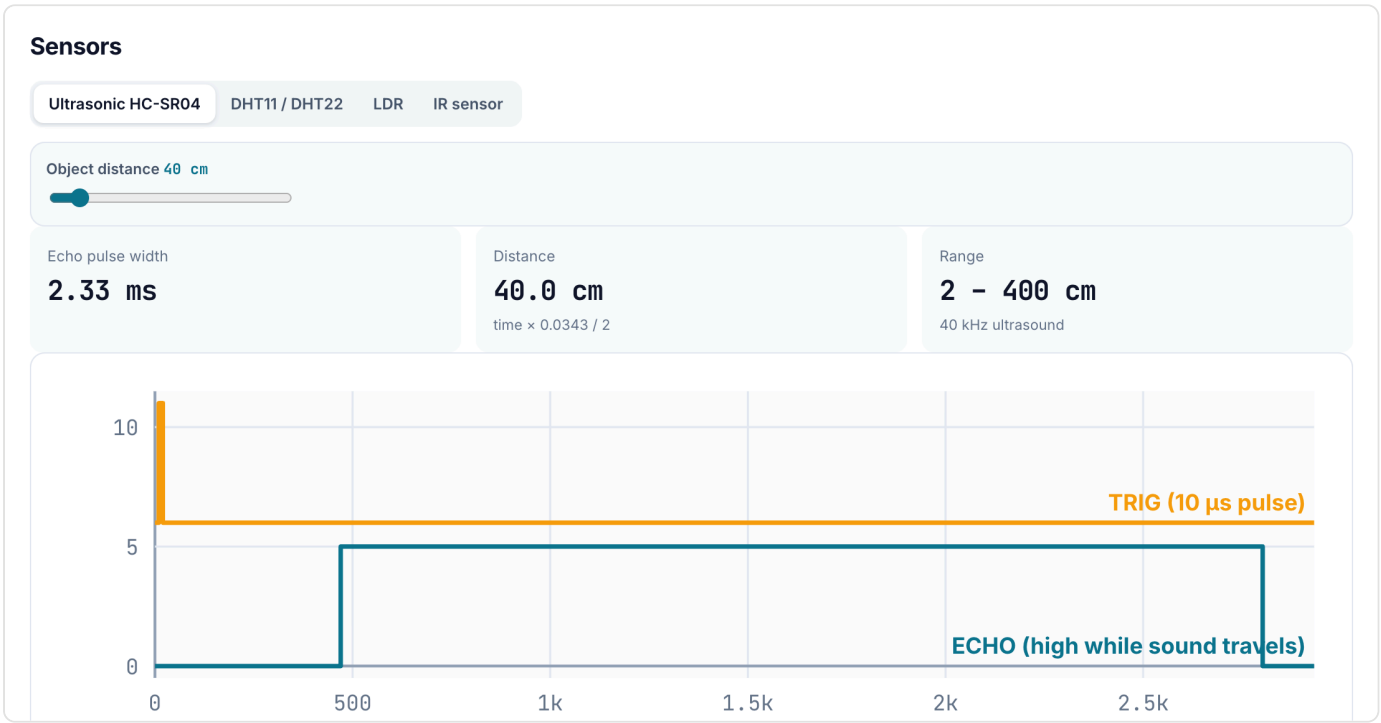


Fig. 35 · **Sensors.** The HC-SR04 tab: trigger and echo pulses for the chosen distance, the distance formula, the sketch and the serial monitor output. The DHT tab shows the 40-bit packet with its checksum; LDR and IR tabs show the divider and digital output.

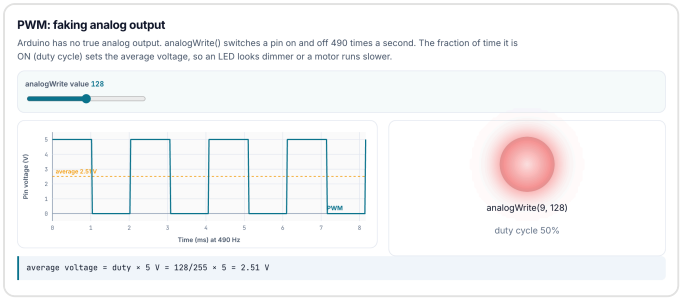


Fig. 36 · **PWM.** Duty cycle and the average voltage it gives, analogWrite values 0–255.

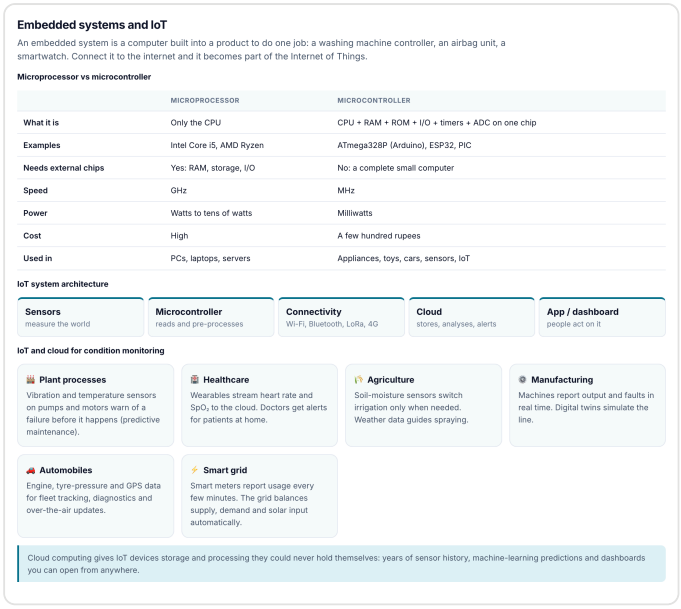
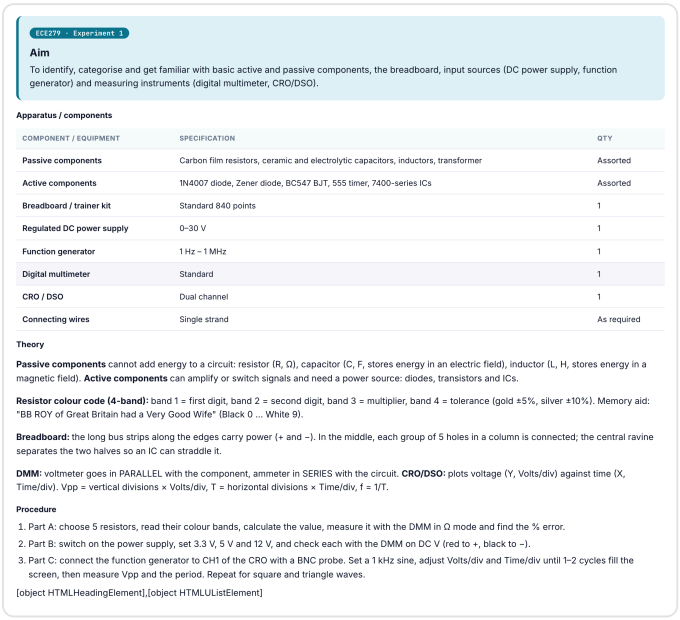


Fig. 37 · **Embedded systems and IoT.** Microprocessor vs microcontroller, and IoT uses in industry, healthcare, agriculture, vehicles and the smart grid.

09 · LAB (ECE279)

Seven experiments on a virtual bench

ECE131 has no practical hours of its own, so the hands-on part follows the ECE279 lab that goes with the course: the KVL and diode experiments support Units I and IV, and the Arduino experiment supports Unit VI. Every experiment follows the format of a lab file. **Aim and theory** gives the aim, apparatus, theory, procedure and precautions. **Virtual bench** is where the readings are taken. **Observations and result** fills the tables from those readings and writes the result. **Viva questions** lists questions asked in the viva, with answers.



ECE279 · Experiment 1

Aim

To identify, categorise and get familiar with basic active and passive components, the breadboard, input sources (DC power supply, function generator) and measuring instruments (digital multimeter, CRO/DSO).

Apparatus / components

COMPONENT / EQUIPMENT	SPECIFICATION	QTY
Passive components	Carbon film resistors, ceramic and electrolytic capacitors, inductors, transformer	Assorted
Active components	1N4007 diode, Zener diode, BC547 BJT, 555 timer, 7400-series ICs	Assorted
Breadboard / trainer kit	Standard 840 points	1
Regulated DC power supply	0–30 V	1
Function generator	1 Hz – 1 MHz	1
Digital multimeter	Standard	1
CRO / DSO	Dual channel	1
Connecting wires	Single strand	As required

Theory

Passive components cannot add energy to a circuit: resistor (R, Ω), capacitor (C, F, stores energy in an electric field), inductor (L, H, stores energy in a magnetic field). **Active components** can amplify or switch signals and need a power source: diodes, transistors and ICs.

Resistor colour code (4-band): band 1 = first digit, band 2 = second digit, band 3 = multiplier, band 4 = tolerance (gold ±5%, silver ±10%). Memory aid: "BB ROY of Great Britain had a Very Good Wife" (Black 0 … White 9).

Breadboard: the long bus strips along the edges carry power (+ and −). In the middle, each group of 5 holes in a column is connected; the central ravine separates the two halves so an IC can straddle it.

DMM: voltmeter goes in PARALLEL with the component, ammeter in SERIES with the circuit. **CRO/DSO:** plots voltage (Y, Volts/div) against time (X, Time/div). Vpp = vertical divisions × Volts/div, T = horizontal divisions × Time/div, f = 1/T.

Procedure

1. Part A: choose 5 resistors, read their colour bands, calculate the value, measure it with the DMM in Ω mode and find the % error.
2. Part B: switch on the power supply, set 3.3 V, 5 V and 12 V, and check each with the DMM on DC V (red to +, black to −).
3. Part C: connect the function generator to CH1 of the CRO with a BNC probe. Set a 1 kHz sine, adjust Volts/div and Time/div until 1–2 cycles fill the screen, then measure Vpp and the period. Repeat for square and triangle waves.

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Fig. 38 · **Aim and theory tab** of Experiment 1.



Questions commonly asked in the viva for this experiment. Click to reveal the answer.

▸ **Q1.** What is the advantage of a JK over an SR flip-flop?

▾ **Q2.** What is the race-around condition?

With J = K = 1, if the clock pulse is wider than the propagation delay, the output toggles repeatedly within a single clock pulse.

▸ **Q3.** How is race-around removed?

▸ **Q4.** Why is the 7410 used here?

▸ **Q5.** How do you make a T flip-flop from a JK?

▸ **Q6.** How do you make a D flip-flop from a JK?

Fig. 39 · **Viva tab** of Experiment 6, one answer opened.

EXP 1 Introduction to components and instruments

A component gallery (passive and active), a colour-code reader where you choose the bands and "measure" the resistor (the reading has a realistic tolerance error), a breadboard where clicking a hole highlights everything connected to it, a regulated supply with a DMM, and a function generator connected to a CRO with volts/div, time/div and Autoset. Readings go into three tables.

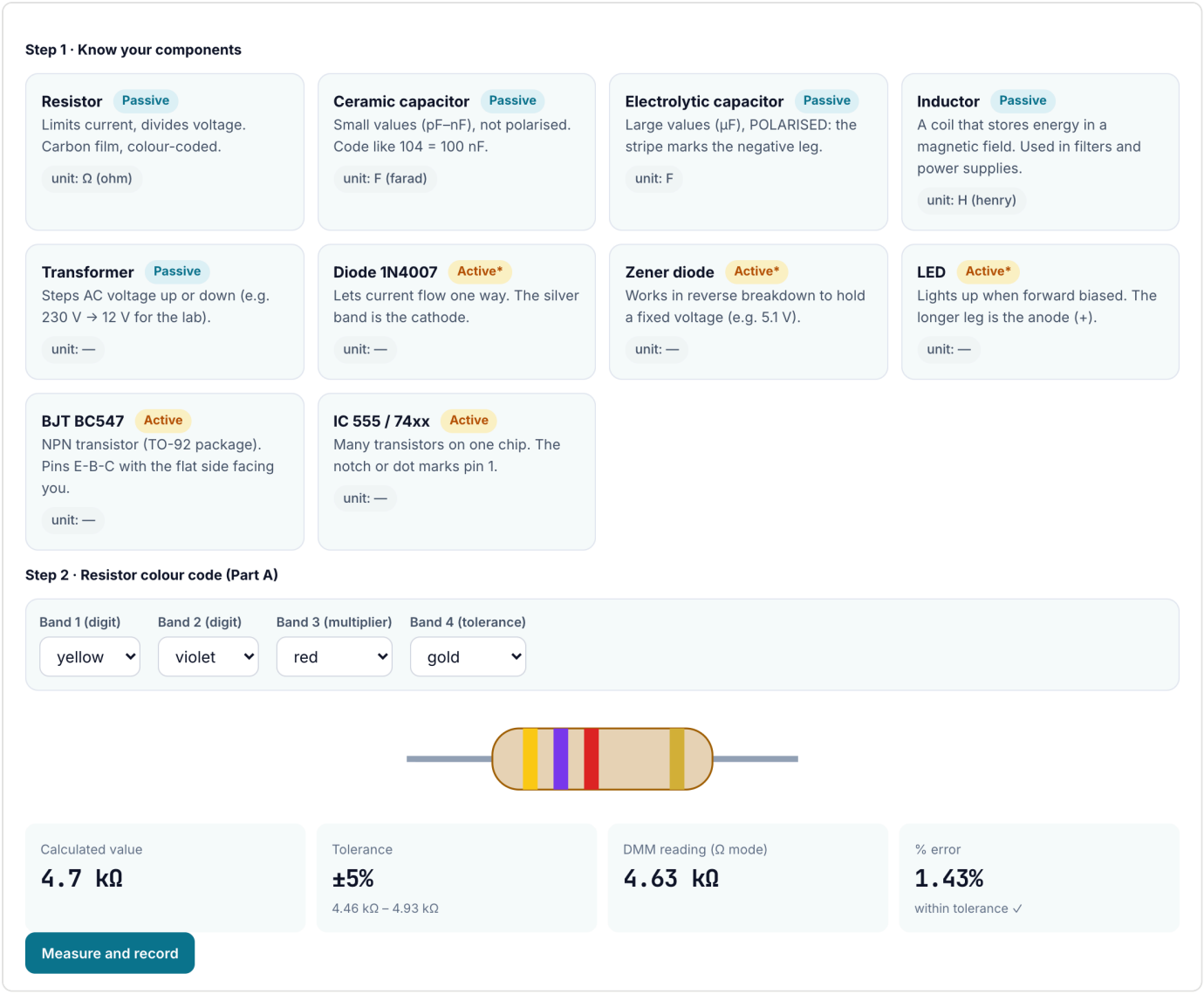


Fig. 40 · **Experiment 1 bench.** Colour code, breadboard, supply and DMM, function generator and CRO.

EXP 2 KVL voltage hunt

A T-network on a 10 V supply: R1 in series with R2 ∥ R3, values chosen from E12. The actual resistors differ from their marked values by up to ±4%, and the meter has a little noise, so the measured loop sums come out close to zero but not exactly, like in the real lab. Place the red and black probes on nodes to measure any voltage.

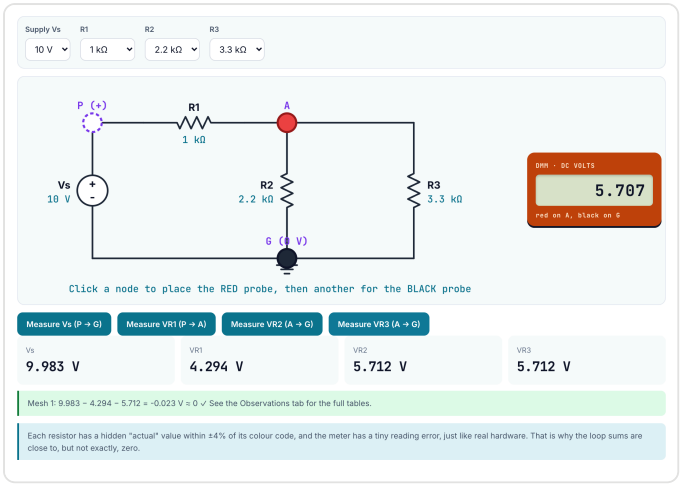


Fig. 41 · **Bench.** Probes on the nodes, the DMM reading and the four measurements.

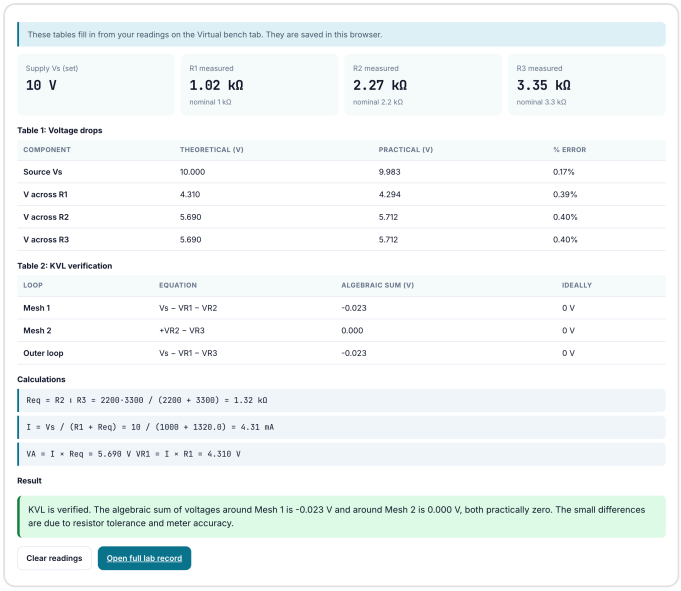


Fig. 42 · **Observations.** Voltage drops (theory vs measured) and the loop sums.

EXP 3 Diode V-I characteristics (1N4007)

The supply drives the diode through 100 Ω. The diode voltage and current are the solution of the Shockley equation with the loop equation, found by bisection. Readings can be taken one at a time or as a full set; the graph, knee voltage, static and dynamic resistance are worked out from the recorded points.

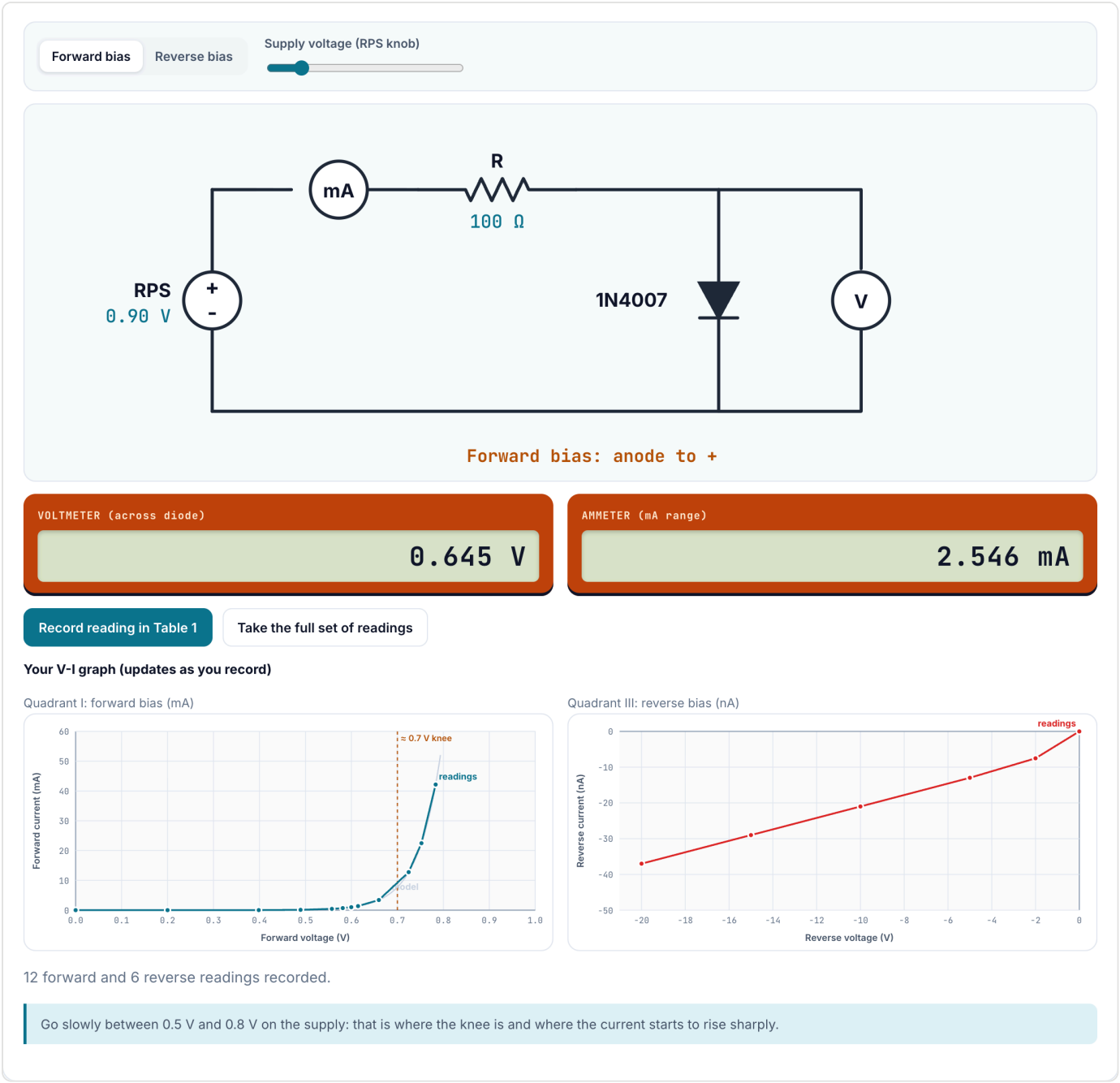


Fig. 43 · **Bench.** Forward-bias circuit with voltmeter and milliammeter, and the V-I graph drawn from the readings: forward in quadrant I (mA) and reverse in quadrant III (nA).

These tables fill in from your readings on the Virtual bench tab. They are saved in this browser.

Table 1: Forward bias (1N4007)

#	SOURCE VOLTAGE (V)	DIODE VOLTAGE (V)	DIODE CURRENT (MA)
1	0.00	0.000	0.000
2	0.20	0.200	0.000
3	0.40	0.398	0.017
4	0.50	0.489	0.106
5	0.60	0.557	0.425
6	0.65	0.581	0.688
7	0.70	0.600	1.003
8	0.75	0.614	1.356
9	1.00	0.660	3.403
10	2.00	0.725	12.755
11	3.00	0.752	22.476
12	5.00	0.783	42.167

Table 2: Reverse bias

#	SOURCE VOLTAGE (V)	DIODE VOLTAGE (V)	REVERSE CURRENT (MA)
1	0.0	0.00	0.000
2	2.0	2.00	0.008
3	5.0	5.00	0.013
4	10.0	10.00	0.021
5	15.0	15.00	0.029
6	20.0	20.00	0.037

Fig. 44 · **Observations and result.** Forward and reverse tables, the graph and the calculations, all from the recorded readings.

EXP 4 Universal gates (7400, 7402)

NOT, AND and OR built from NAND only and from NOR only. For each circuit the IC pinout is drawn with the pins you wire highlighted and labelled, next to a connection table and the live gate circuit. "Record this row" saves the LED state for the current switches; the observation table marks each row ✓ or ✗ against the expected value.

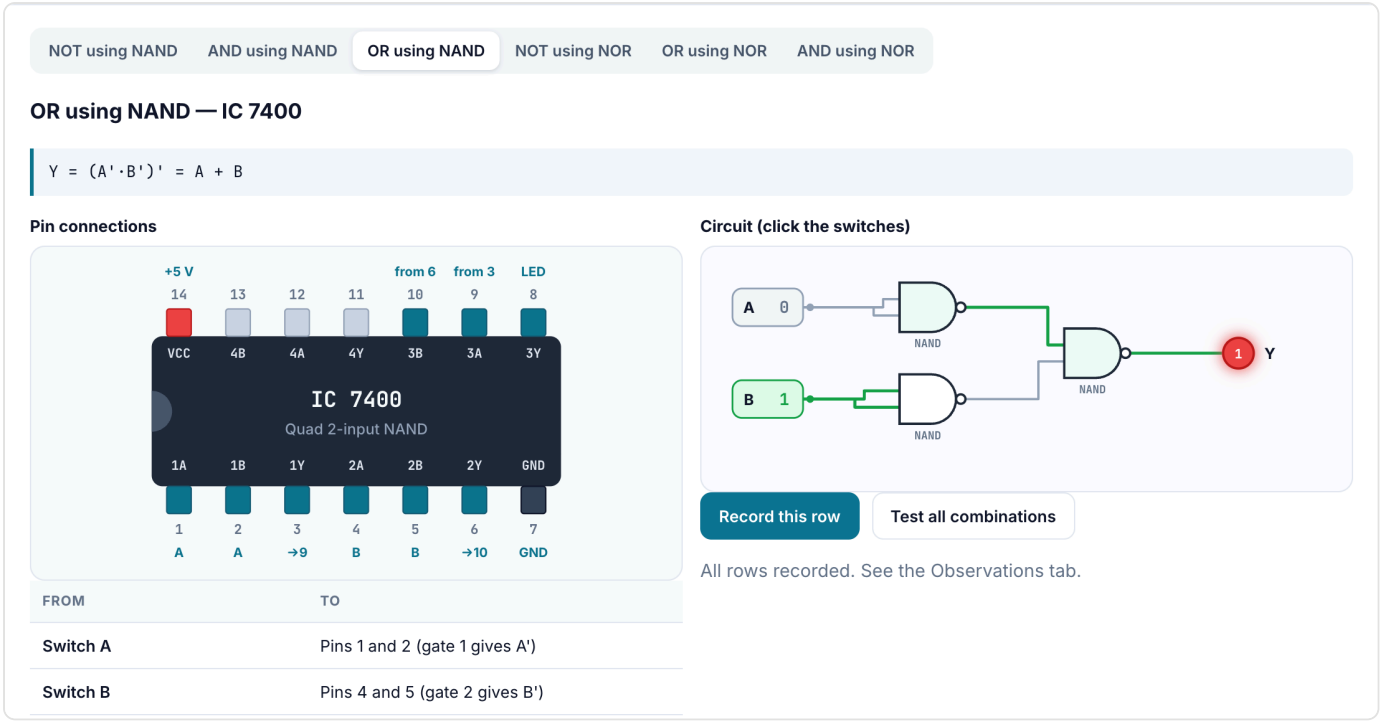


Fig. 45 · **Bench.** OR using NAND: A to pins 1 and 2, B to pins 4 and 5, pins 3 and 6 to pins 9 and 10, pin 8 to the LED. Note that the 7402 has its outputs on pins 1, 4, 10 and 13, which the drawing shows when you switch to the NOR circuits.

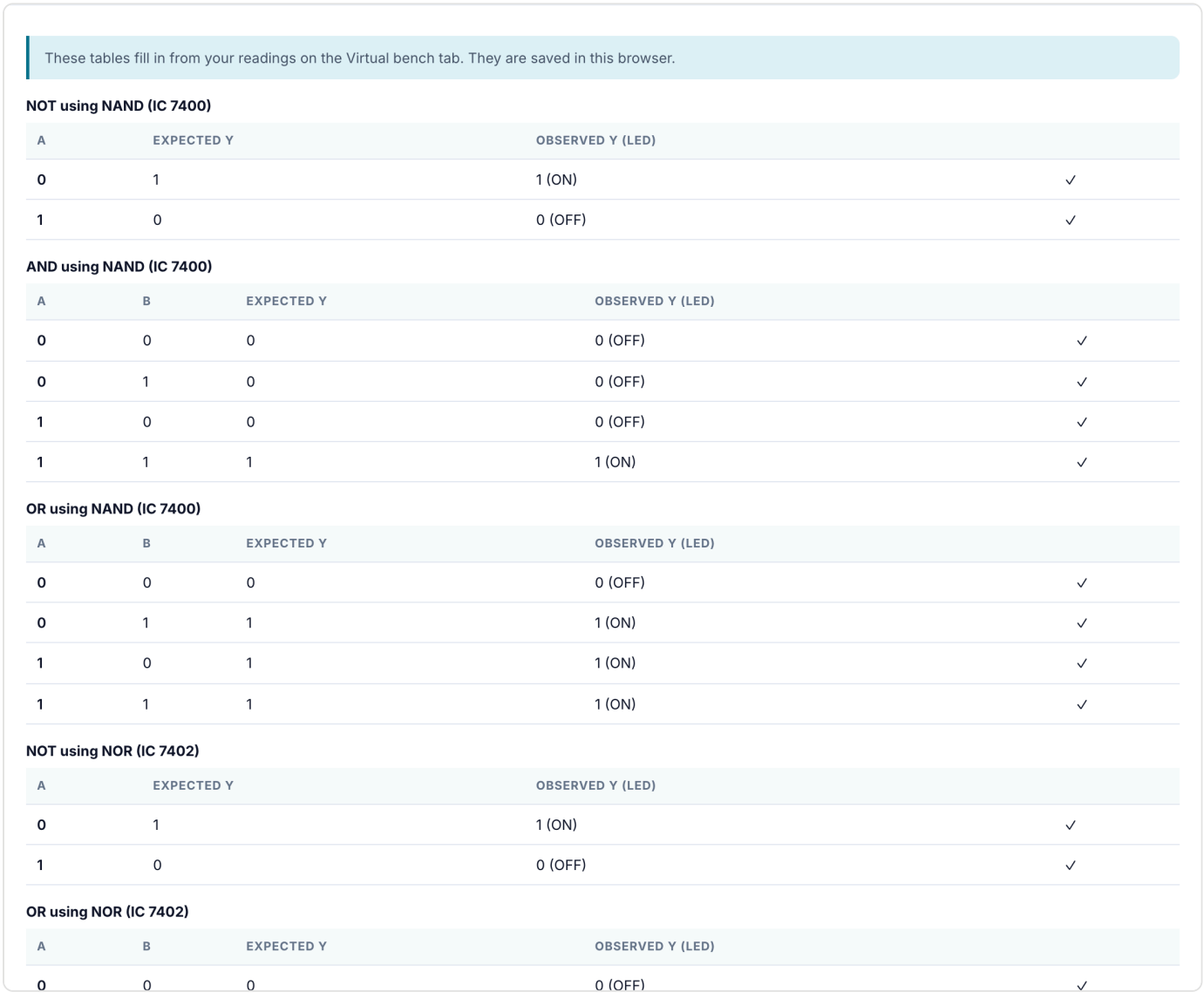


Fig. 46 · **Observations.** All six circuits verified.

EXP 5 Half and full adder (7486, 7408, 7432)

The pin wiring of all three chips for the full adder: gate 1 of the 7486 makes A⊕B, gate 2 adds Cin; the two AND gates of the 7408 make AB and Cin(A⊕B); the 7432 ORs them into Cout.

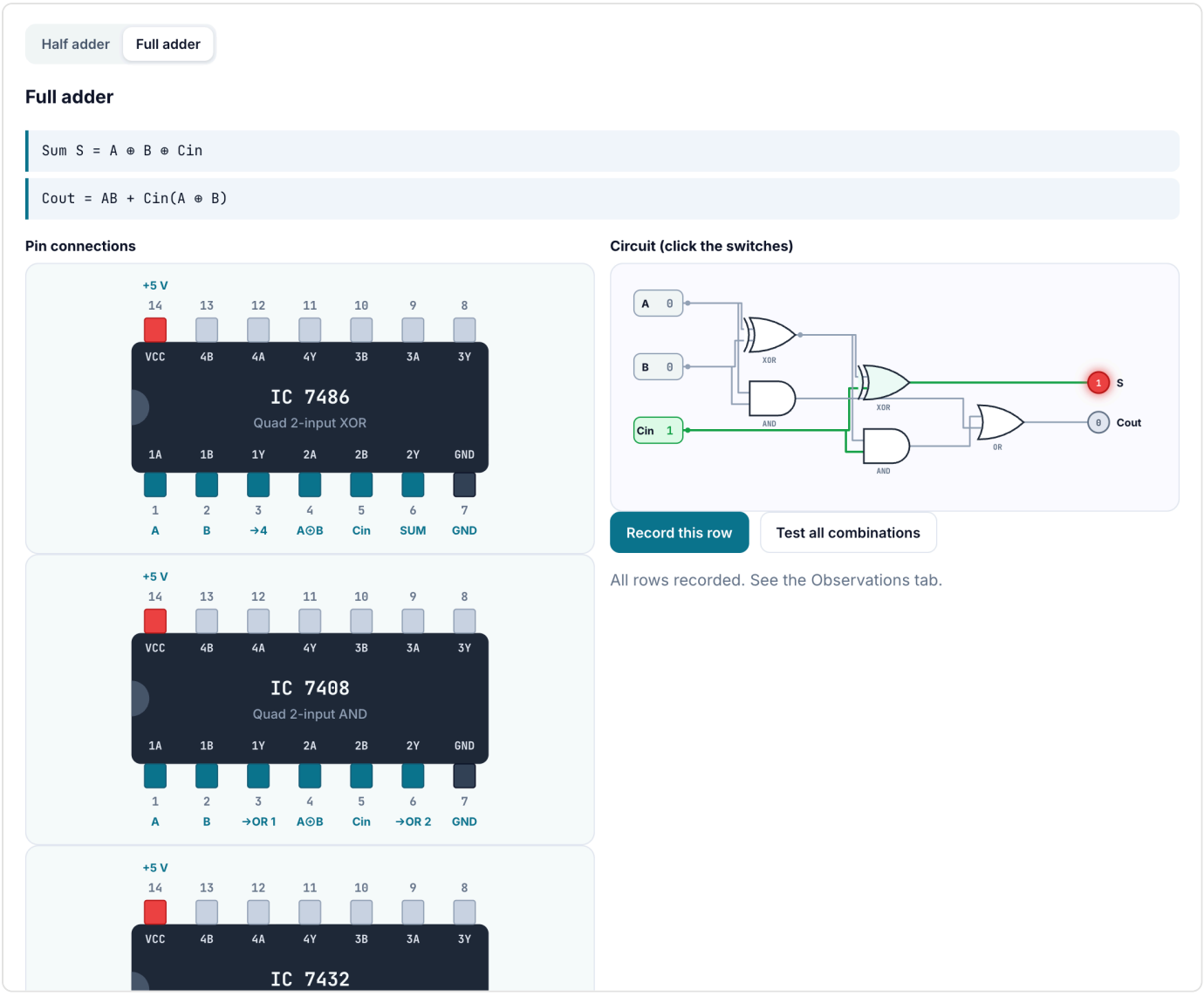
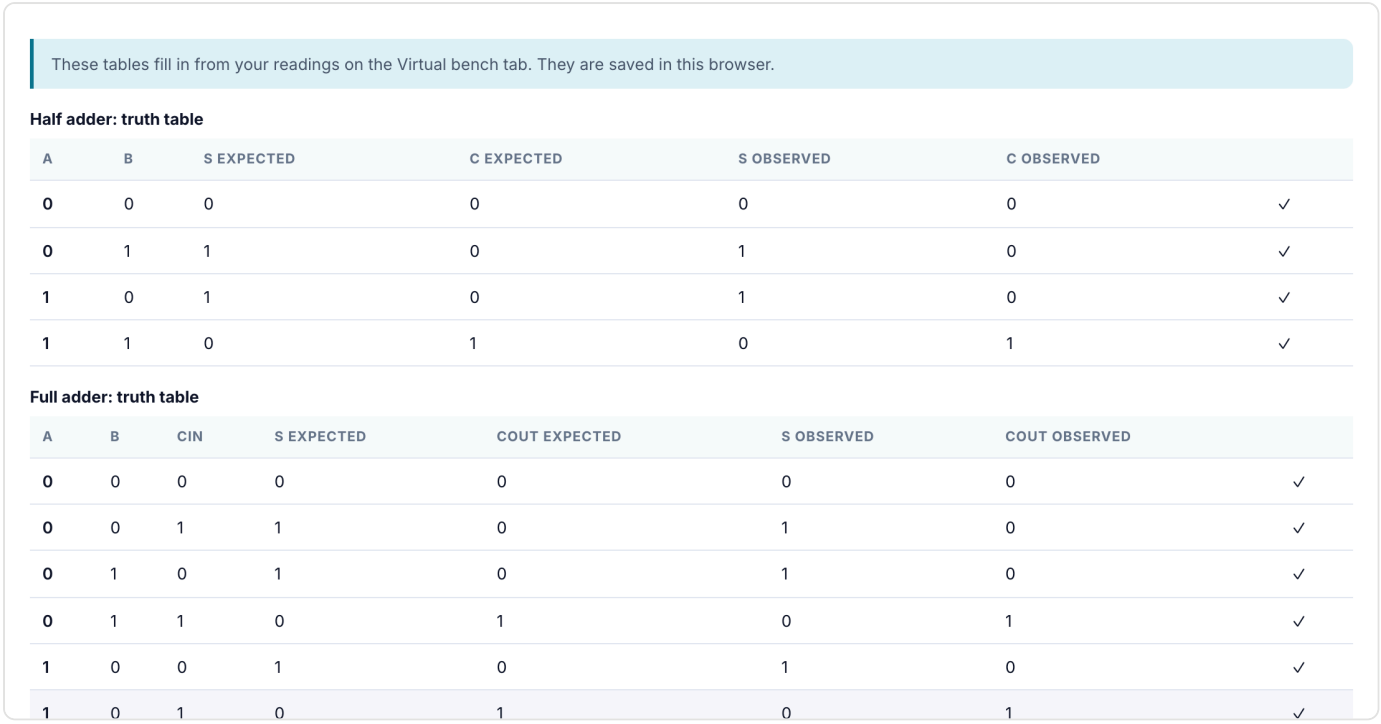


Fig. 47 · **Bench.** Full adder pinouts and live circuit.



These tables fill in from your readings on the Virtual bench tab. They are saved in this browser.

Half adder: truth table

A	B	S EXPECTED	C EXPECTED	S OBSERVED	C OBSERVED	
0	0	0	0	0	0	✓
0	1	1	0	1	0	✓
1	0	1	0	1	0	✓
1	1	0	1	0	1	✓

Full adder: truth table

A	B	CIN	S EXPECTED	COUT EXPECTED	S OBSERVED	COUT OBSERVED	
0	0	0	0	0	0	0	✓
0	0	1	1	0	1	0	✓
0	1	0	1	0	1	0	✓
0	1	1	0	1	0	1	✓
1	0	0	1	0	1	0	✓
1	0	1	0	1	0	1	✓

Fig. 48 · **Observations.** Both truth tables verified, observed against expected.

EXP 6 JK flip-flop using NAND gates (7400, 7410)

The steering gates are 3-input NANDs from the 7410 (J, CLK, Q' and K, CLK, Q), and the latch is two NANDs of the 7400. A **short clock pulse** (two gate delays) lets the flip-flop change exactly once, so the truth table can be recorded: hold, reset, set, toggle. **Holding the clock high** with J = K = 1 shows the race-around condition and records how many times Q changed.

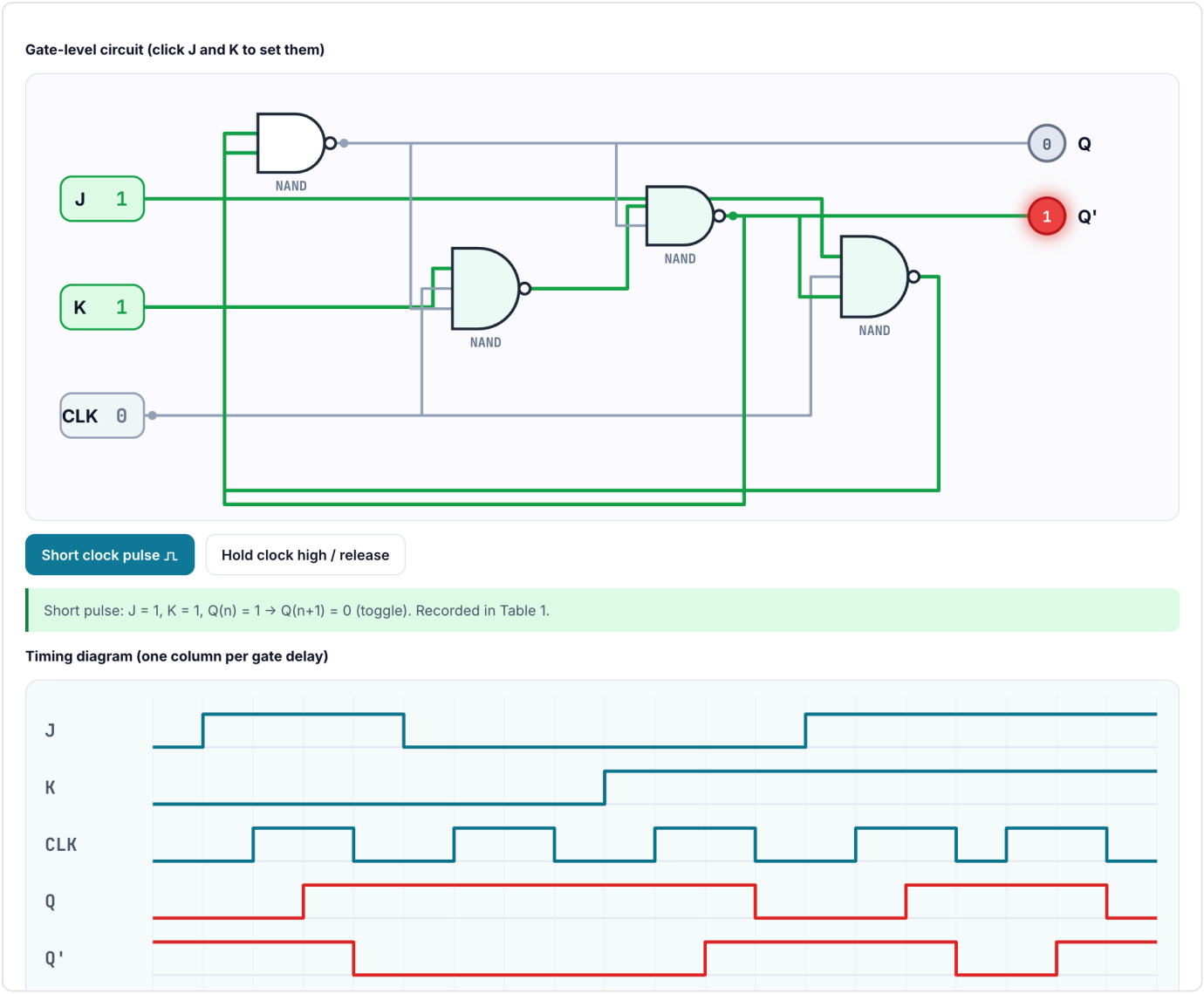


Fig. 49 · **Bench.** Gate-level circuit, a timing diagram with one column per gate delay after a sequence of set, hold, reset and two toggles. The pin connections of both ICs are drawn below it on the same tab.

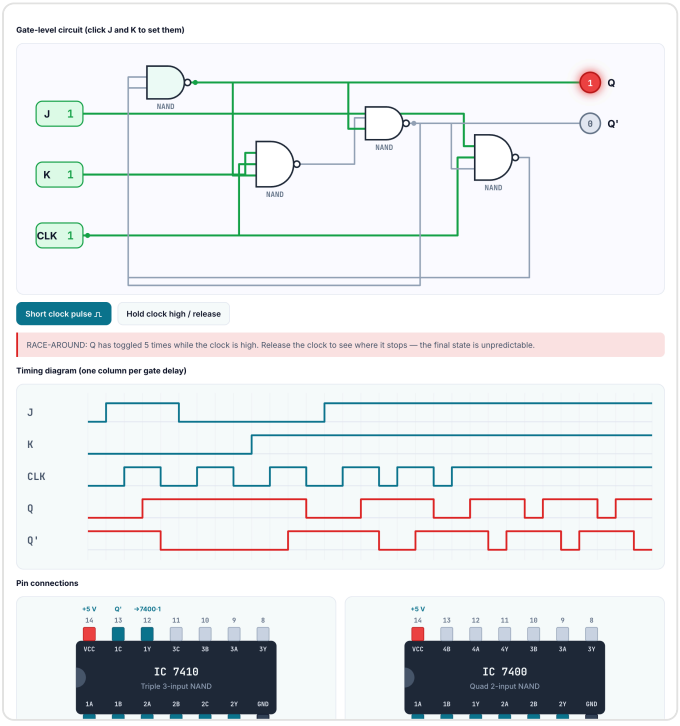


Fig. 50 · **Race-around.** Clock held high with J = K = 1: Q oscillates.

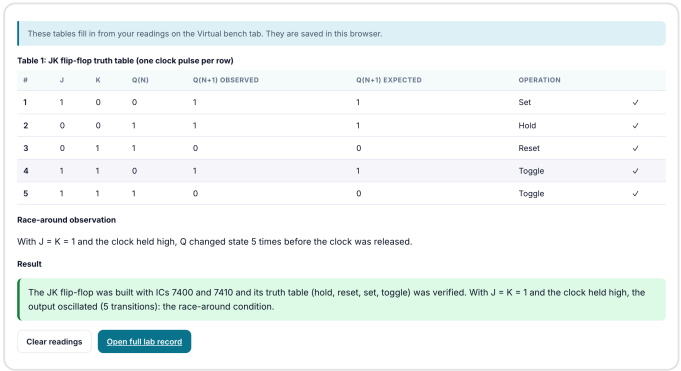


Fig. 51 · **Observations.** The truth table from the recorded pulses and the race-around observation.

EXP 7 Arduino projects

Three sense → decide → act projects with the full sketch, the connection table, a block diagram, the serial monitor and a picture that follows the same logic as the code: an **automatic street light** (LDR on A0, threshold 300, relay on pin 7, status LED on 13), a **temperature-controlled vent** (DHT11 on pin 2, servo on pin 5 opening to 90° above 28 °C) and a **parking distance alarm** (HC-SR04 trig 9 / echo 10, buzzer on pin 8 below 20 cm).

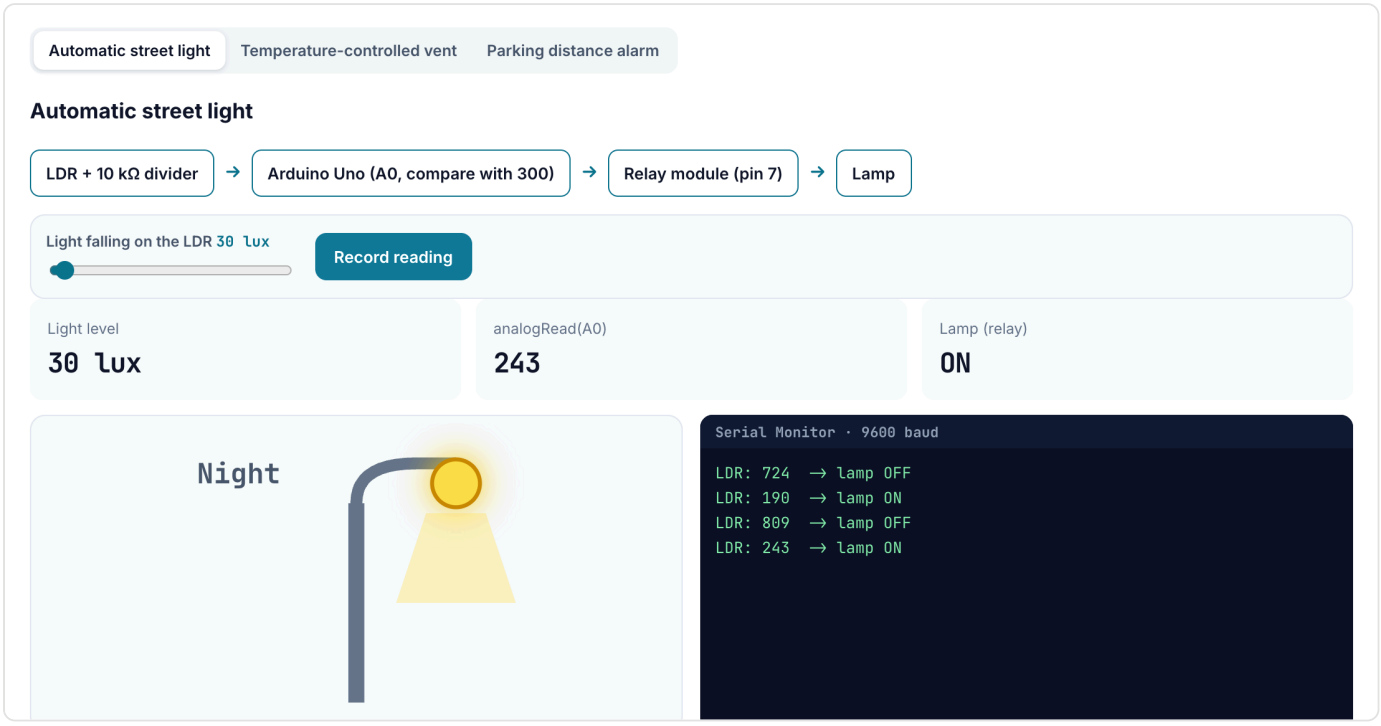


Fig. 52 · **Street light.** In the dark the ADC value falls below 300 and the lamp switches on.

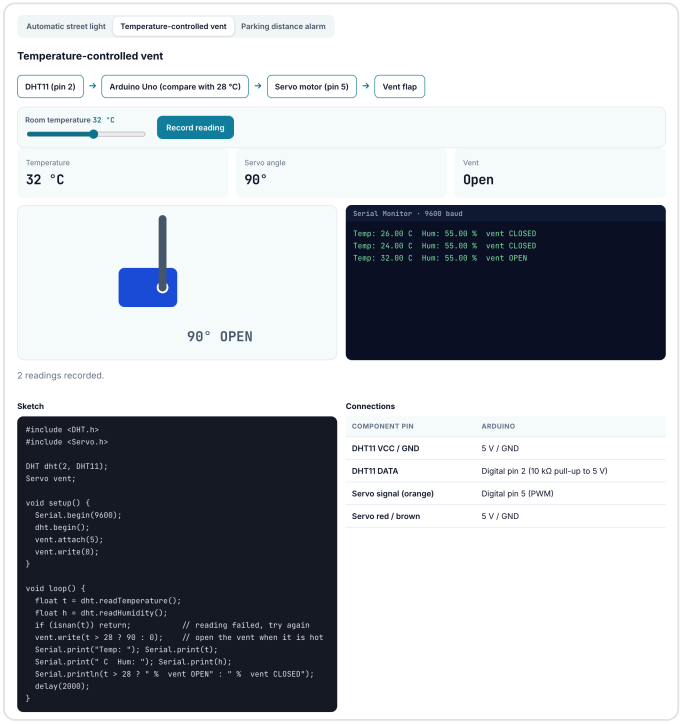


Fig. 53 · **Temperature vent.** 32 °C: the servo turns to 90°.

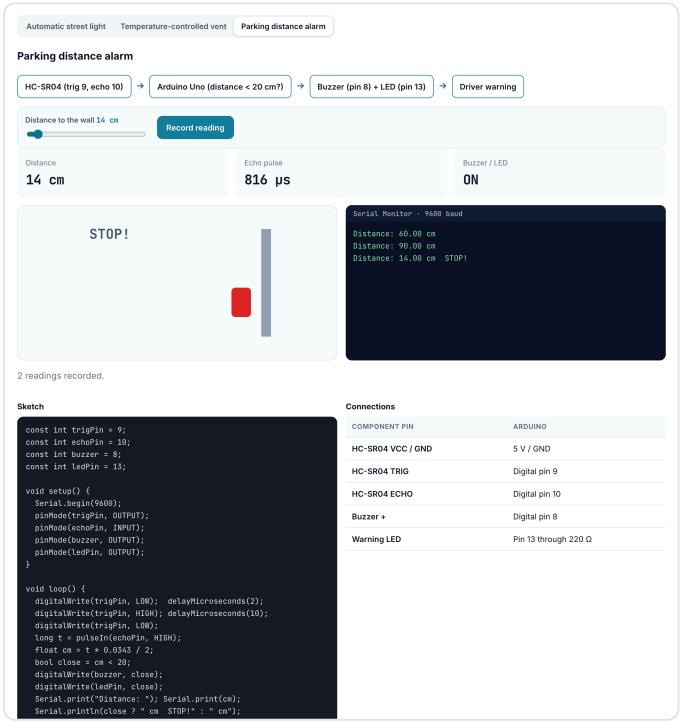
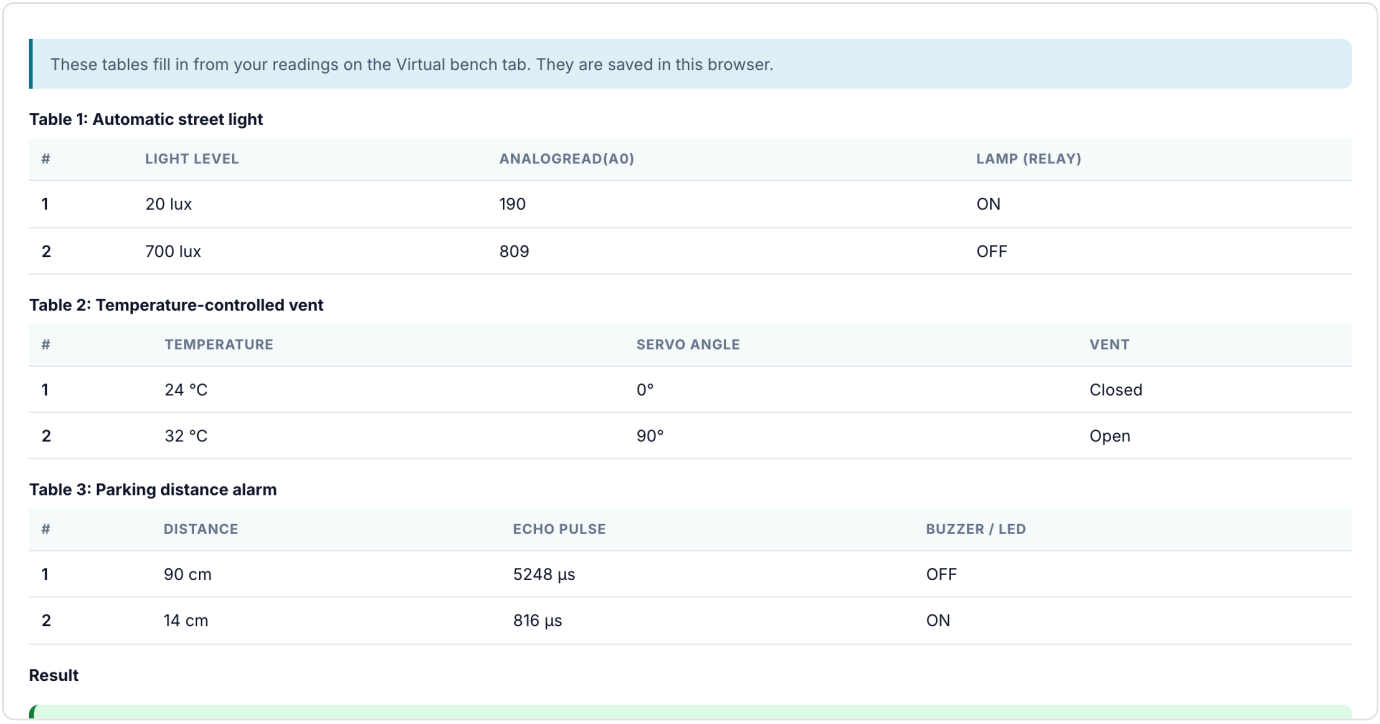


Fig. 54 · **Parking alarm.** 14 cm: buzzer and LED on.



These tables fill in from your readings on the Virtual bench tab. They are saved in this browser.

Table 1: Automatic street light

#	LIGHT LEVEL	ANALOGREAD(A0)	LAMP (RELAY)
1	20 lux	190	ON
2	700 lux	809	OFF

Table 2: Temperature-controlled vent

#	TEMPERATURE	SERVO ANGLE	VENT
1	24 °C	0°	Closed
2	32 °C	90°	Open

Table 3: Parking distance alarm

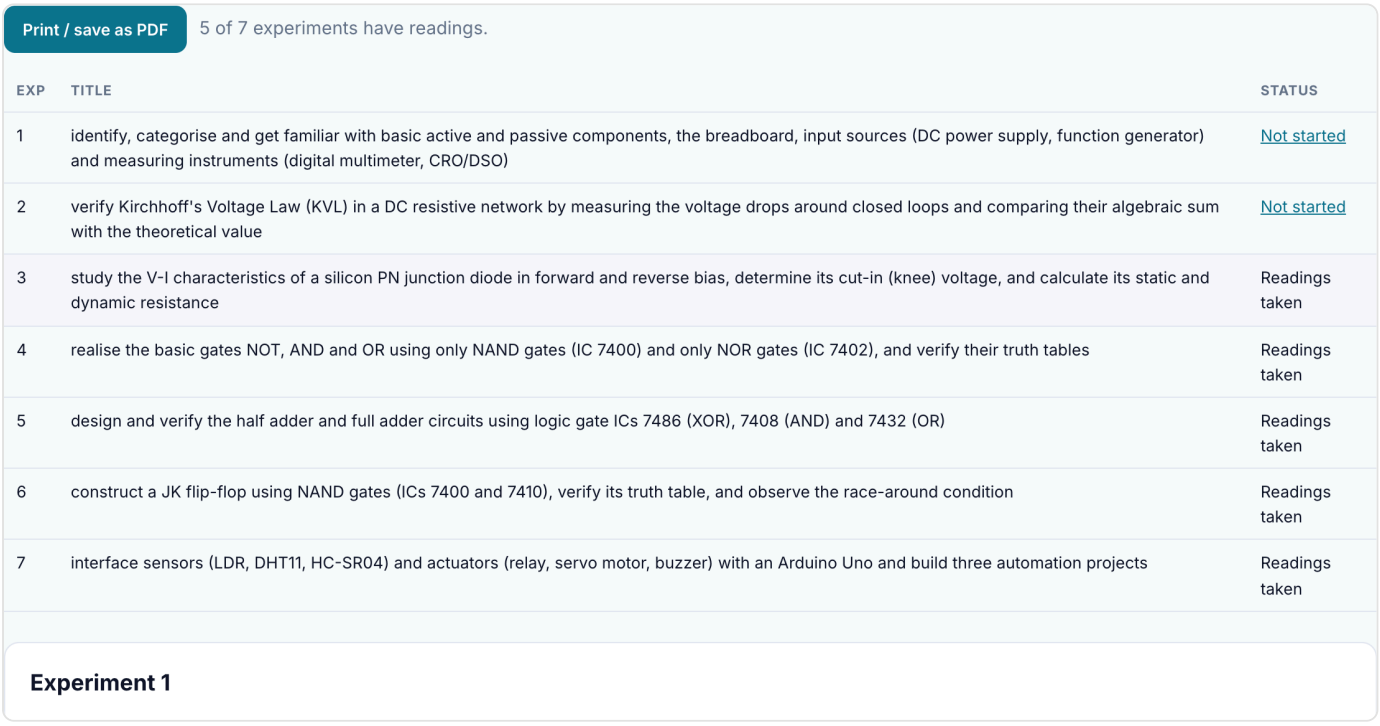
#	DISTANCE	ECHO PULSE	BUZZER / LED
1	90 cm	5248 µs	OFF
2	14 cm	816 µs	ON

Result

Fig. 55 · **Observations.** Readings on both sides of each threshold.

Lab record

The Lab record page puts the aim, apparatus, observation tables and result of all seven experiments into one document. It has print styles, so "Print / save as PDF" gives a clean lab file without the sidebar.



Print / save as PDF

5 of 7 experiments have readings.

EXP	TITLE	STATUS
1	identify, categorise and get familiar with basic active and passive components, the breadboard, input sources (DC power supply, function generator) and measuring instruments (digital multimeter, CRO/DSO)	Not started
2	verify Kirchhoff's Voltage Law (KVL) in a DC resistive network by measuring the voltage drops around closed loops and comparing their algebraic sum with the theoretical value	Not started
3	study the V-I characteristics of a silicon PN junction diode in forward and reverse bias, determine its cut-in (knee) voltage, and calculate its static and dynamic resistance	Readings taken
4	realise the basic gates NOT, AND and OR using only NAND gates (IC 7400) and only NOR gates (IC 7402), and verify their truth tables	Readings taken
5	design and verify the half adder and full adder circuits using logic gate ICs 7486 (XOR), 7408 (AND) and 7432 (OR)	Readings taken
6	construct a JK flip-flop using NAND gates (ICs 7400 and 7410), verify its truth table, and observe the race-around condition	Readings taken
7	interface sensors (LDR, DHT11, HC-SR04) and actuators (relay, servo motor, buzzer) with an Arduino Uno and build three automation projects	Readings taken

Experiment 1

Fig. 56 · **Lab record** (top part): the index shows which experiments have readings, followed by each experiment's aim, apparatus, tables and result.

Digital electronics, also covered

The same course was earlier offered as ECE249. Its Units 1 and 2 (electrical laws, semiconductors, BJT, Arduino and sensors) are already shown in chapters 04, 07 and 08. Units 3 to 6 go further into digital electronics than ECE131 does. They are built in ElectroBench as well and are also the theory behind Experiments 4, 5 and 6.

ECE249 UNIT	TOPICS	IN THIS REPORT
Unit 1	Ohm, KCL, KVL, dividers, semiconductors, PN diode, rectifiers, BJT	Chapters 04 and 07
Unit 2	Analog and digital signals, Arduino Uno, HC-SR04, DHT11/22, LDR, IR	Chapter 08
Unit 3	Number systems, complements, codes, gates, Boolean laws, SOP/POS, K-maps	This chapter
Unit 4	Adders, subtractors, MUX, DEMUX, decoder, encoder, comparators	This chapter
Units 5 and 6	Latches, flip-flops, race-around, conversions, shift registers, counters	This chapter

Unit 3 · Number systems, logic gates and K-maps

Unit 3 covers binary, octal, decimal and hexadecimal conversion, 1's and 2's complements, BCD, Excess-3 and Gray code, logic gates, universal gates, Boolean laws, SOP and POS forms and Karnaugh maps.

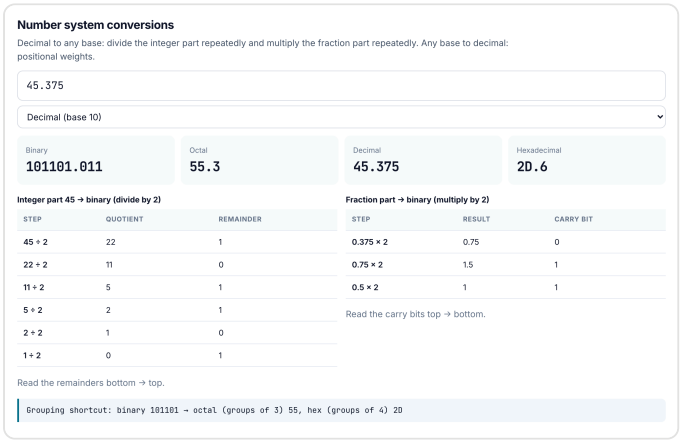


Fig. 57 · **Conversions.** Every step of the division or multiplication method is shown, including fractions.

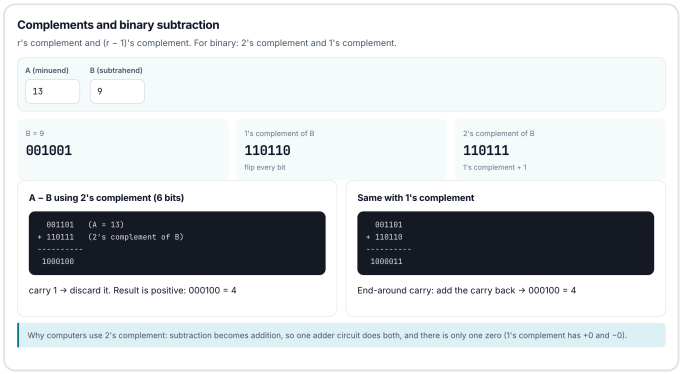


Fig. 58 · **Complements.** Subtraction with 1's complement (end-around carry) and 2's complement.

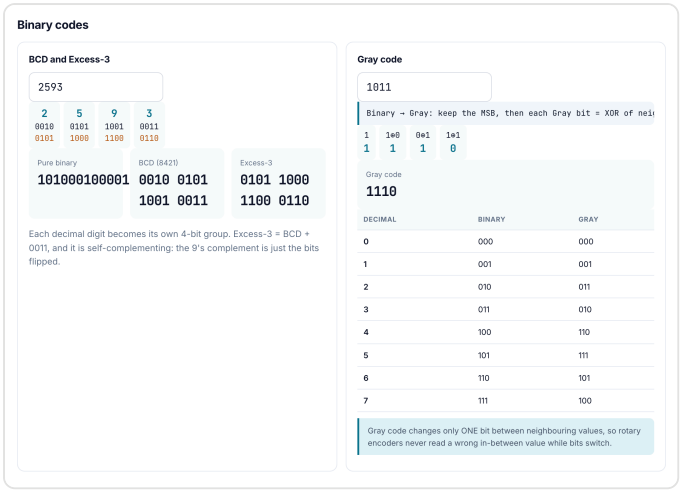


Fig. 59 · **Codes.** BCD, Excess-3 and Gray code for any number, with the XOR steps for Gray.

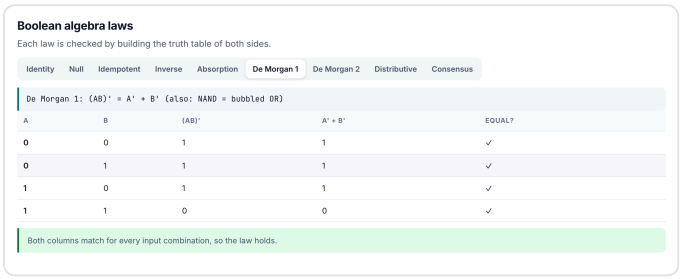


Fig. 61 · **Boolean laws.** Type any expression: the truth table, canonical SOP and POS are generated.

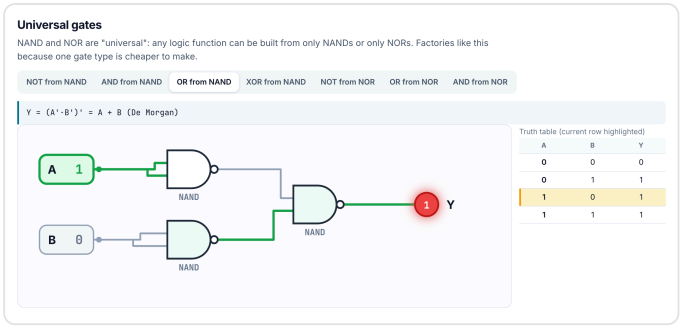


Fig. 60 · **Universal gates.** OR built only from NAND gates. Click the switches: green wires carry logic 1.

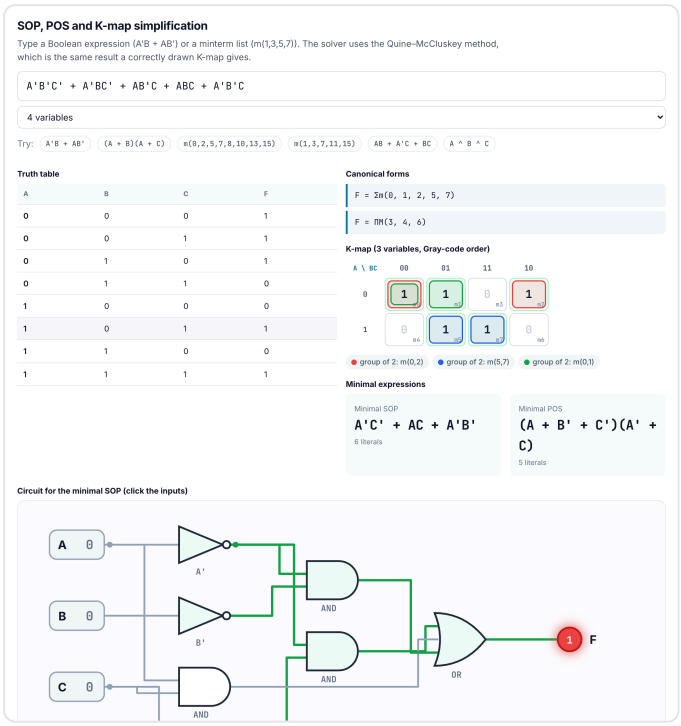


Fig. 62 · **K-map solver.** Click cells (1, 0 or don't care). The map is minimised with the Quine–McCluskey method, the groups are drawn, and the minimised circuit is built and simulated.

Combinational circuits

Unit 4 covers half and full adders and subtractors, the 4:1 multiplexer, 1:4 demultiplexer, 2:4 decoder, 4:2 encoder and 1-bit and 2-bit comparators. Each one here is a gate-level netlist simulated by the same engine.

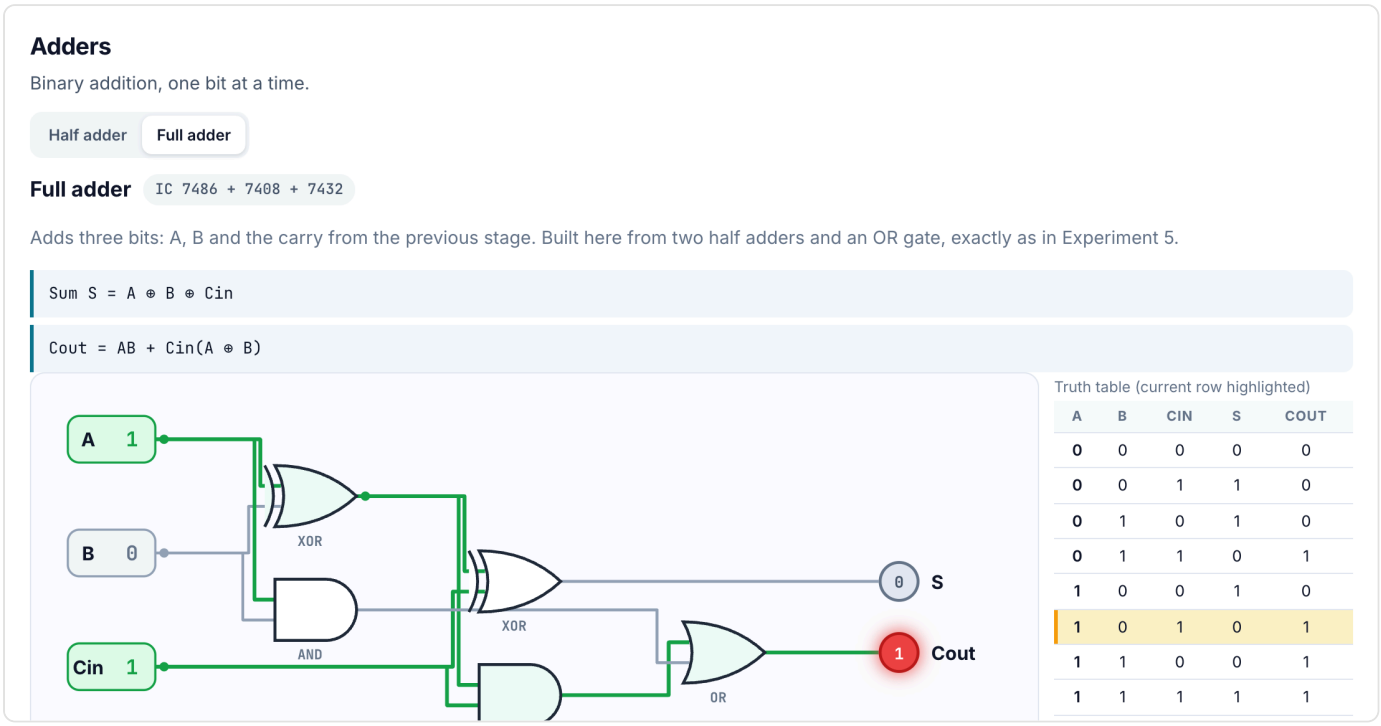


Fig. 63 · **Full adder.** Built from two half adders and an OR gate, exactly as wired in Experiment 5. Here A = 1, B = 0, Cin = 1 gives S = 0, Cout = 1.

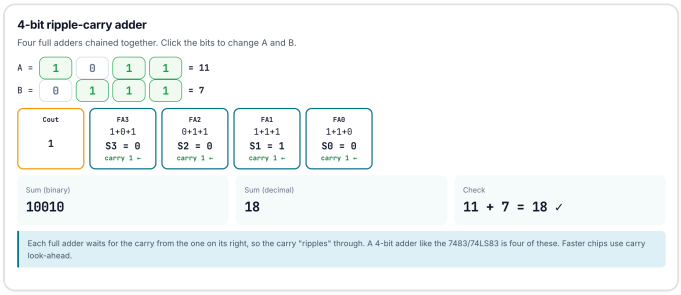


Fig. 64 · **4-bit ripple-carry adder.** Four full adders chained, carry rippling from stage to stage.

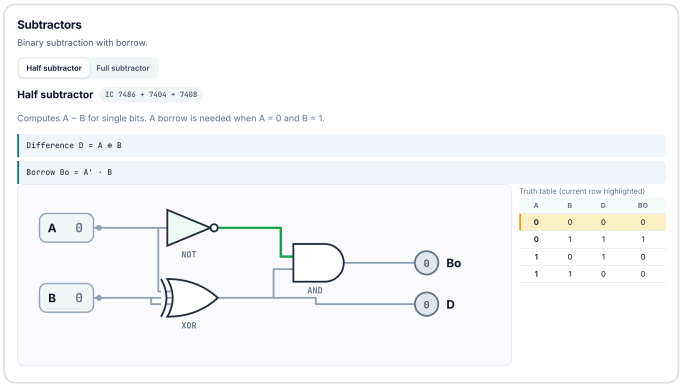


Fig. 65 · **Subtractors.** Difference and borrow circuits.

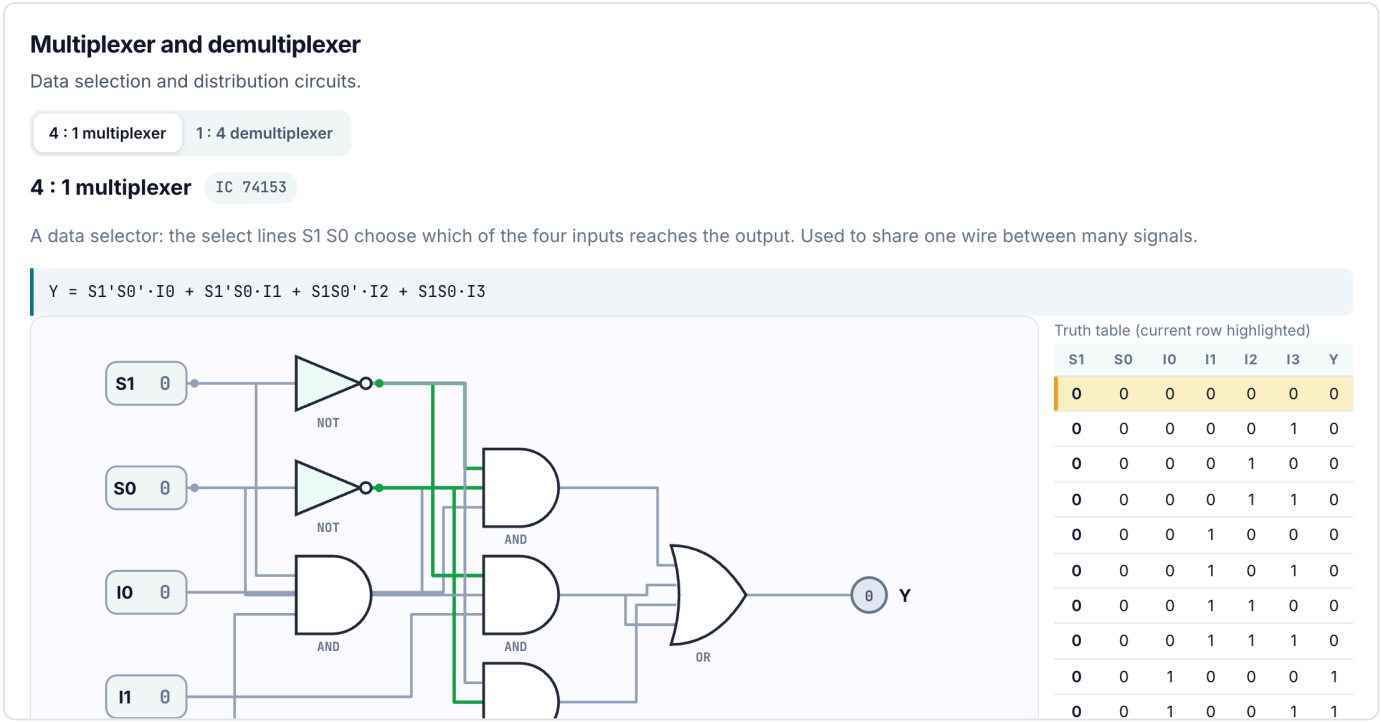


Fig. 66 · **MUX and DEMUX.** The select lines route one of four inputs to the output.

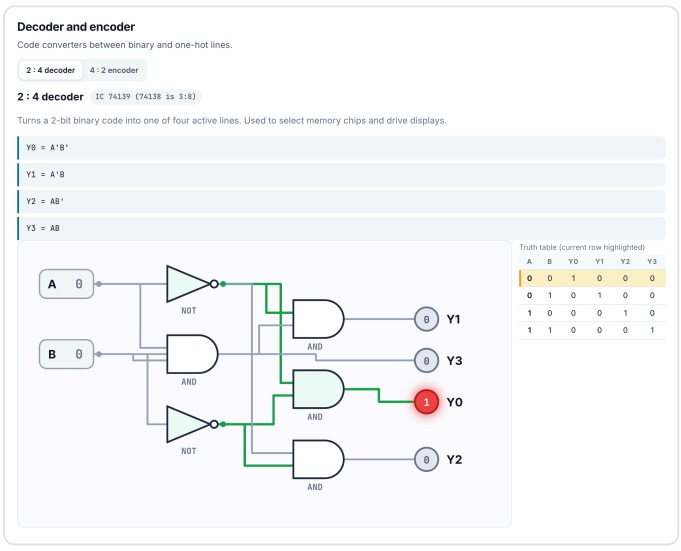


Fig. 67 · **Decoder and encoder.** 2:4 decoder and 4:2 encoder.

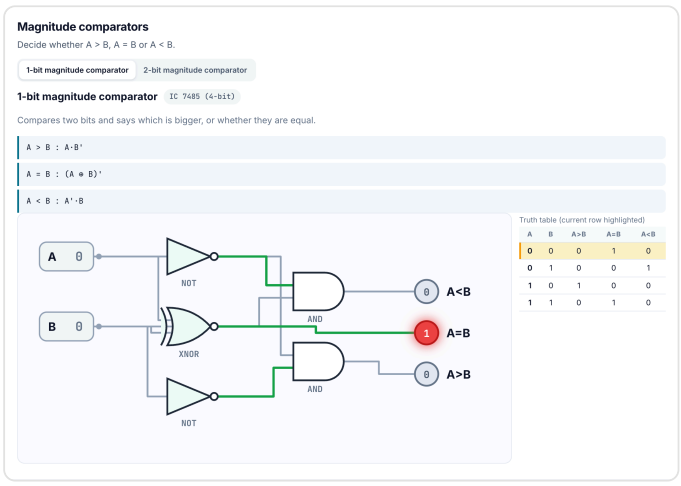


Fig. 68 · **Comparators.** A > B, A = B and A < B outputs for 1-bit and 2-bit numbers.

Flip-flops, registers and counters

Unit 5 covers SR latches with NOR and NAND gates, the D latch, SR, JK, D and T flip-flops, the race-around condition, the master–slave JK and flip-flop conversions. Unit 6 covers shift registers and counters.

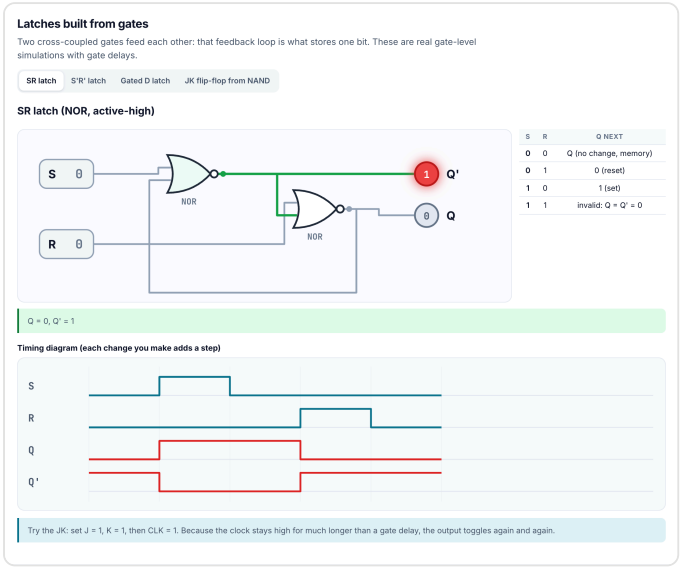


Fig. 69 · **SR latch from NOR gates.** Two cross-coupled gates with real gate delays. The timing diagram records each change.

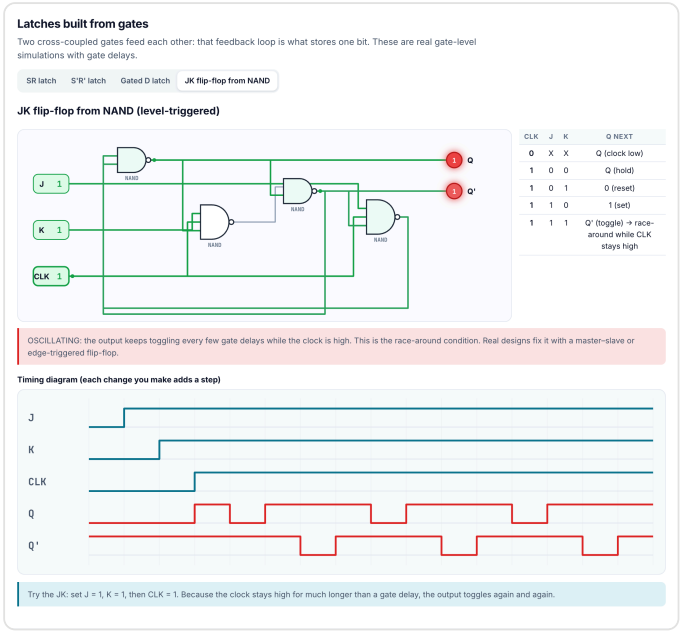


Fig. 70 · **Race-around.** JK from NAND gates with J = K = 1 and the clock held high: Q keeps toggling, gate delay by gate delay.

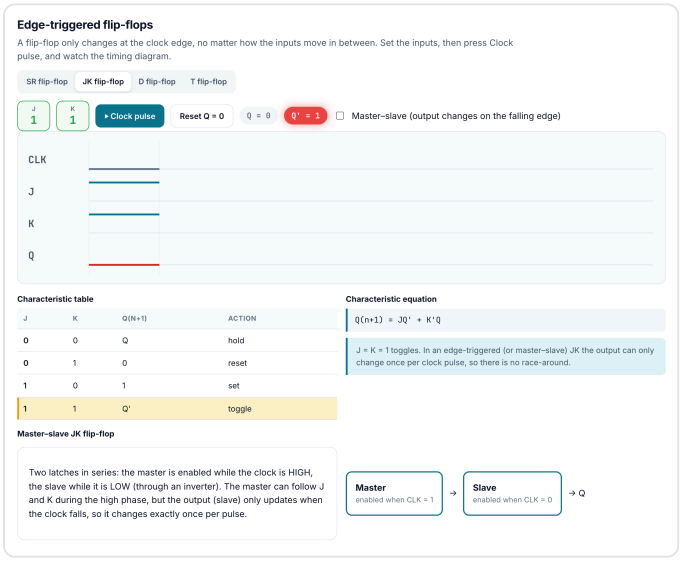


Fig. 71 · **Flip-flops.** Edge-triggered SR, JK, D and T with timing diagrams, and a master–slave option.

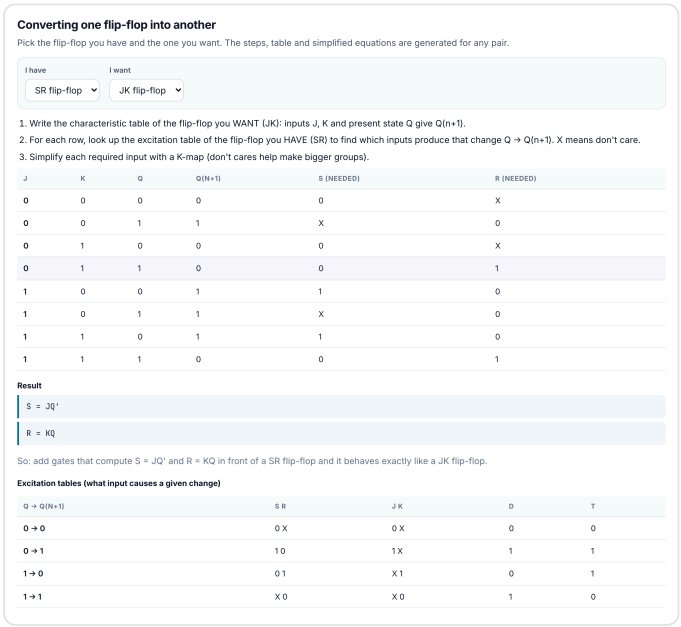


Fig. 72 · **Conversions.** Pick any two flip-flops. The excitation table, K-maps and input equations are worked out (e.g. SR → JK gives S = JQ', R = KQ).

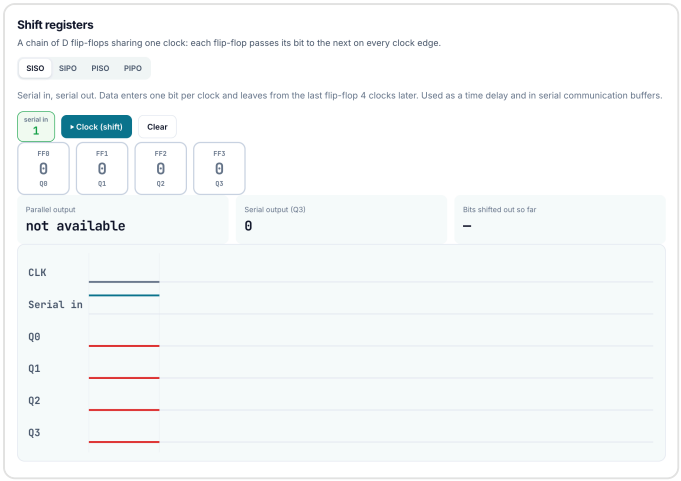


Fig. 73 · **Shift registers.** SISO, SIPO, PISO and PIPO, clocked bit by bit.

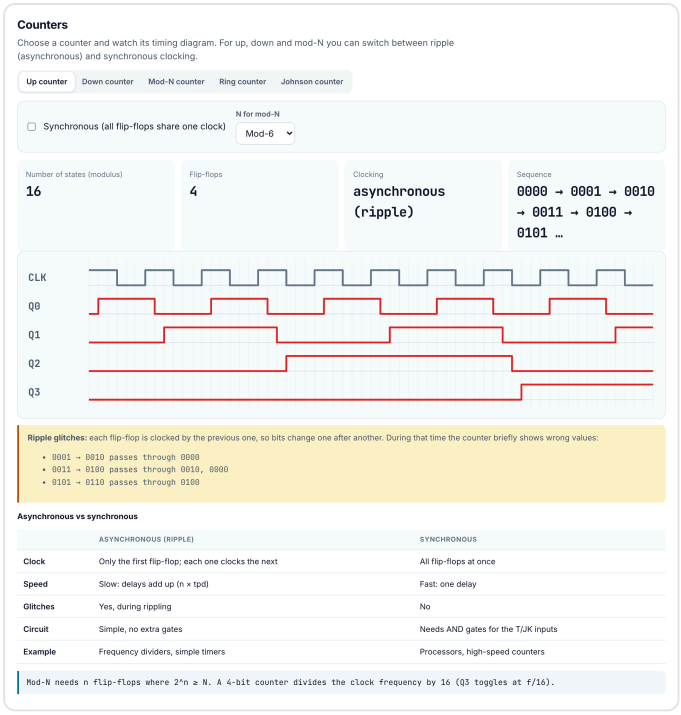


Fig. 74 · **Counters.** Up, down, mod-N, ring and Johnson. Switching to ripple mode shows the glitch states that a synchronous counter avoids.

Every line of the syllabus

Every line of the ECE131 2026-27 syllabus with the place in ElectroBench that covers it and its course outcome, followed by the lab experiments and the earlier ECE249 syllabus.

ECE131 · Basic Electrical and Electronics Engineering (session 2026-27)

SYLLABUS TOPIC	WHERE IN ELECTROBENCH	UNIT / CO
R, L, C, voltage, current, power and energy; Ohm's law; Kirchhoff's laws	DC: Basics, KCL & KVL (Fig. 3)	I · CO1, CO2
Series and parallel simplification, voltage and current division	DC: Series–parallel, Divider rules (Fig. 4)	I · CO2
Star–delta transformation; dependent and independent sources	DC: Star–delta, Mesh & nodal (Fig. 5, 6)	I · CO2
Mesh and nodal analysis	DC: Mesh & nodal (Fig. 6)	I · CO2
Thevenin, Norton, maximum power transfer, superposition	DC: Theorems (Fig. 7)	I · CO2
Amplitude, phase, phase difference, RMS and average value; notation i, v, I, V	AC: Waveform (Fig. 8)	II · CO2
Complex impedance; RL, RC, RLC series; power factor and power	AC: RL · RC · RLC (Fig. 9)	II · CO2
Resonance in series RLC	AC: Resonance (Fig. 10)	II · CO2
Three-phase star and delta; line and phase relations	AC: Three-phase (Fig. 11)	II · CO2
Mutual inductance, transformer working, turns ratio, transformer on DC, auto- and instrument transformers	Machines: Transformer (Fig. 12)	III · CO3
DC machines: principle, classification, starting, speed control, applications	Machines: DC motor (Fig. 13)	III · CO3
Single- and three-phase induction motors; AC motor applications	Machines: Induction motor (Fig. 14)	III · CO3
Fleming's left- and right-hand rules	Machines: Fleming's rules (Fig. 15)	III · CO3
PN junction diode working and characteristics	Semiconductors (Fig. 17, 18); Exp 3	IV · CO4
Zener diode; varactor diode	Semiconductors: Zener, Varactor (Fig. 20, 21)	IV · CO4

SYLLABUS TOPIC	WHERE IN ELECTROBENCH	UNIT / CO
Logic gates	Logic: Gates (Fig. 27); Exp 4	IV · CO4
Testing of diode and BJT	Semiconductors: Switch & testing (Fig. 22)	IV · CO4
Half wave, full wave and bridge rectifiers	Semiconductors: Rectifiers (Fig. 19)	IV · CO4
MOSFET representation and characteristics; applications of diodes, transistors, MOSFET	BJT & MOSFET (Fig. 25, 26)	IV · CO4
Filters: low-, high-, band-pass, band-stop (active)	Op-amps: Filters (Fig. 31)	V · CO5
Op-amp properties, virtual ground	Op-amps: Properties (Fig. 28)	V · CO5
Inverting, non-inverting, adder, subtractor	Op-amps: Configurations (Fig. 29)	V · CO5
Integrator and differentiator	Op-amps: Integrator & differentiator (Fig. 30)	V · CO5
Comparator and its use in anti-lock braking	Op-amps: Comparator & ABS (Fig. 32)	V · CO5
Embedded systems; microprocessor vs microcontroller	Arduino: Embedded systems & IoT (Fig. 37)	VI · CO6
Arduino and sensors	Arduino: Uno, Sensors (Fig. 34, 35); Exp 7	VI · CO6
IoT and cloud in plants, healthcare, agriculture, manufacturing, automobiles, smart grid	Arduino: Embedded systems & IoT (Fig. 37)	VI · CO6

ECE131 course outcomes

CO1
Enumerate the basic terminology of electrical and electronics engineering

DC basics, component gallery (Exp 1), properties tables on every device page

CO2
Understand the behaviour and notation of electrical circuits

DC circuits and AC circuits pages, Exp 2

CO3
Explain the principles and applications of electrical machines

Transformer, DC motor, induction motor, Fleming's rules

CO4
Determine the characteristics of semiconductor devices

Diode, Zener, varactor, BJT, MOSFET pages; Exp 3 measures a real characteristic

CO5
Analyse filters and operational amplifiers

Op-amp configurations, integrator, filters, comparator and ABS

CO6
Illustrate IoT, cloud and embedded systems in various fields

Arduino page and the three projects of Exp 7

Lab · ECE279

PRACTICAL	VIRTUAL BENCH IN ELECTROBENCH	ECE131 / ECE249 UNIT
1. Introduction to components and instruments	Gallery, colour code, breadboard, RPS + DMM, function generator + CRO (Fig. 40)	I / 1
2. KVL voltage hunt	T-network with probes and loop sums (Fig. 41, 42)	I / 1
3. Diode V-I characteristics	1N4007 forward and reverse, knee, static and dynamic resistance (Fig. 43, 44)	IV / 1
4. Universal gates	7400 and 7402 pin wiring, six circuits (Fig. 45, 46)	IV / 3
5. Half and full adder	7486 + 7408 + 7432 pin wiring (Fig. 47, 48)	– / 4
6. JK flip-flop using NAND gates	7400 + 7410, short pulse vs held clock (Fig. 49, 50, 51)	– / 5
7. Arduino projects with sensors	Street light, temperature vent, parking alarm (Fig. 52, 53, 54)	VI / 2

Earlier syllabus · ECE249

SYLLABUS TOPIC	WHERE IN ELECTROBENCH	UNIT
Ohm's law, KCL, KVL, voltage and current divider rules	DC circuits: Basics, KCL & KVL, Divider rules (Fig. 3, 4); Exp 2	Unit 1
Intrinsic and extrinsic semiconductors	Semiconductors: Doping (Fig. 16)	Unit 1
PN junction diode, V-I characteristics	Semiconductors: PN junction, V-I curve (Fig. 17, 18); Exp 3	Unit 1
Rectifiers: half wave, full wave, bridge	Semiconductors: Rectifiers (Fig. 19)	Unit 1
Diode as a switch	Semiconductors: Switch & testing (Fig. 22)	Unit 1
BJT construction, types, modes, CE characteristics	BJT & MOSFET: Construction, CE characteristics (Fig. 23, 24)	Unit 1
Analog and digital signals	Arduino: Analog vs digital (Fig. 33)	Unit 2
Arduino Uno board and pins	Arduino: Arduino Uno (Fig. 34)	Unit 2
HC-SR04, DHT11/DHT22, LDR, IR sensors	Arduino: Sensors (Fig. 35); Exp 7	Unit 2
Number systems and conversions	Number systems: Conversions (Fig. 57)	Unit 3
1's and 2's complement	Number systems: Complements (Fig. 58)	Unit 3
BCD, Excess-3, Gray code	Number systems: Codes (Fig. 59)	Unit 3
Logic gates, universal gates	Logic: Gates, Universal gates (Fig. 27, 60); Exp 4	Unit 3
Boolean laws, SOP and POS	Logic: Boolean laws (Fig. 61)	Unit 3

SYLLABUS TOPIC	WHERE IN ELECTROBENCH	UNIT
Karnaugh maps	Logic: K-map solver (Fig. 62)	Unit 3
Half and full adder, half and full subtractor	Combinational: Adders, 4-bit adder, Subtractors (Fig. 63, 64, 65); Exp 5	Unit 4
MUX 4:1, DEMUX 1:4	Combinational: MUX & DEMUX (Fig. 66)	Unit 4
Decoder 2:4, encoder 4:2	Combinational: Decoder & encoder (Fig. 67)	Unit 4
1-bit and 2-bit comparators	Combinational: Comparators (Fig. 68)	Unit 4
SR latch (NOR, NAND), D latch	Latches & flip-flops: Latches (Fig. 69)	Unit 5
SR, JK, D, T flip-flops	Latches & flip-flops: Flip-flops (Fig. 71); Exp 6	Unit 5
Race-around condition, master–slave JK	Latches (Fig. 70), Flip-flops; Exp 6 (Fig. 50)	Unit 5
Flip-flop conversions	Latches & flip-flops: Conversions (Fig. 72)	Unit 5
Shift registers SISO, SIPO, PISO, PIPO	Registers & counters: Shift registers (Fig. 73)	Unit 6
Asynchronous and synchronous counters, up/down, mod-N, ring, Johnson	Registers & counters: Counters (Fig. 74)	Unit 6

ECE249 · Basic Electrical and Electronics Engineering (earlier syllabus)

The earlier version of the course, with more digital electronics. Each row is a topic from its unit notes and the place in ElectroBench where it is built.

Unit 1 · Electrical laws and semiconductors

SYLLABUS TOPIC	WHERE	WHAT YOU CAN DO THERE
Ohm's law, KCL, KVL	DC · Basics · DC · KCL & KVL	Solve a two-source network and watch currents balance at a node and voltages sum to zero around loops
Voltage and current divider rules	DC · Divider rules	Change R1, R2 and the source; the divided voltage and currents update
Intrinsic and extrinsic semiconductors	Semiconductors · Doping	Silicon lattice with N-type and P-type doping, majority and minority carriers
PN junction diode, V-I characteristics	Semiconductors · PN junction · V-I curve · Exp 3	Depletion region in forward and reverse bias; Shockley-equation V-I curve
Rectifiers (half wave, full wave, bridge)	Semiconductors · Rectifiers	Simulated output with and without a filter capacitor, ripple and DC level
Diode as a switch	Semiconductors · Switch & testing	ON/OFF states and the ideal-diode model
BJT construction, types, modes, CE characteristics	BJT · Construction & modes · CE characteristics	NPN/PNP, four operating modes, output curves with load line and Q-point

Unit 2 · Arduino and sensors

SYLLABUS TOPIC	WHERE	WHAT YOU CAN DO THERE
Analog and digital signals	Arduino · Analog vs digital	Sampling and quantisation of a sine wave by a 10-bit ADC
Arduino Uno board and pins	Arduino · Arduino Uno	Click any pin of the board to see what it does
Ultrasonic sensor HC-SR04	Arduino · Sensors · Exp 7	Trigger/echo timing, distance formula, sketch and serial output
DHT11 / DHT22	Arduino · Sensors · Exp 7	The 40-bit data packet with checksum, DHT11 vs DHT22
LDR and IR sensors	Arduino · Sensors · Exp 7	LDR divider to ADC value; IR obstacle detection

Unit 3 · Number systems and logic

Fig. 75 · **The ECE249 tab of the course map** in the site: every row links to the page and tab.

How the important parts work

Gate-level simulation with delays

logic.js · lines 47-57

```
step() {
    const next = {};
    let changed = false;
    for (const g of this.gates) {
        const v = OPS[g.type](g.in.map(sig ⇒
this.get(sig))) ? 1 : 0;
        next[g.id] = v;
        if (v ≠≠ this.values[g.id]) changed = true;
    }
    Object.assign(this.values, next);
    return changed;
}
```

Every gate computes its new output from the values of the *previous* instant, then all outputs change together. That is one gate delay.

Combinational circuits settle after a few steps. A latch settles too, because its feedback loop reaches a stable state. A JK flip-flop with J = K = 1 and the clock high never settles: each step flips the output, which feeds back and flips it again. `settle()` reports that, and the page shows it as the race-around condition. Nothing about race-around is hard-coded; it comes out of the gate delays.

A clean clock pulse for Experiment 6

lab6.js · lines 23-27

```
function settleLatch(c) {
    if (c.settle().stable) return;
    for (let k = 0; k < 4; k++) c.gates.forEach(g ⇒ {
c.values[g.id] = OPS[g.type](g.in.map(sig ⇒
c.get(sig))) ? 1 : 0; });
    c.settle();
}
```

A short pulse holds the clock high for two gate delays: long enough for the steering gate and the latch to change once, too short for the new output to come back round. Tests (chapter 13) confirm every J, K, Q combination gives the right next state.

If both latch outputs end up equal, a unit-delay model would oscillate for ever. Real gates are never exactly equal in speed, so the tie is broken with a few in-order passes, the same way the simulator does when a circuit is first built.

The diode operating point

lab3.js · lines 17-30

```
export function diodeReading(Vs, reverse) {
    if (reverse) {
        const I = (0.005 * (1 - Math.exp(-Vs)) + 0.0016
* Vs) * 1e-6; // nA–µA leakage
        return { Vd: Math.max(0, Vs - I * R), I };
    }
    let lo = 0;
    let hi = Math.max(0, Vs);
    for (let k = 0; k < 80; k++) {
        const mid = (lo + hi) / 2;
        if (mid + IS * (Math.exp(mid / (N * VT)) - 1) *
R > Vs) hi = mid; else lo = mid;
    }
    const Vd = (lo + hi) / 2;
    return { Vd, I: (Vs - Vd) / R };
}
```

The current through the series resistor must equal the diode current from the Shockley equation. There is no closed-form answer, so the diode voltage is found by bisection: 80 halvings of the interval 0 to Vs, which is far more precise than the meter shows.

The ideality factor and saturation current ($n \approx 1.9$, $I_S = 5$ nA) put the knee of the simulated 1N4007 at about 0.6–0.7 V, like the real diode.

Nodal analysis for the two-source network

dc.js · lines 13–19

```
export function solveNetwork({ V1, V2, R1, R2, R3 }) {
    const VA = (V1 / R1 + V2 / R3) / (1 / R1 + 1 / R2 +
1 / R3);
    const I1 = (V1 - VA) / R1; // left source → node A
    const I3 = (V2 - VA) / R3; // right source → node A
    const I2 = VA / R2; // node A → ground
    return { VA, I1, I2, I3 };
}
```

Node A is the only unknown, so KCL at A gives one equation: the currents in from both sources equal the current down through R2. Solving it gives VA, and every branch current follows from Ohm's law.

The page then adds up the currents at the node and the voltages around each loop and prints the sums, which is how it demonstrates KCL and KVL.

Series AC circuits

ac.js · lines 65–74

```
export function seriesAC({ type, R, L, C, V, f }) {
    const w = TAU * f;
    const XL = type.includes('L') ? w * L : 0;
    const XC = type.includes('C') ? 1 / (w * C) : 0;
    const X = XL - XC;
    const Z = Math.hypot(R, X);
    const theta = Math.atan2(X, R);
    const I = V / Z;
    return { XL, XC, X, Z, theta, I, VR: I * R, VL: I *
XL, VC: I * XC, pf: Math.cos(theta), P: V * I *
Math.cos(theta), Q: V * I * Math.sin(theta), S: V * I };
}
```

Reactances come from ω = 2πf. The impedance is the magnitude of R + jX, and the phase angle its argument. Power factor, real, reactive and apparent power follow. The same function drives the phasor diagram and the resonance sweep.

IC pinouts

ic.js · lines 17–35

```
export function drawIC(part, used = {}) {
    const ic = ICS[part];
    const w = 420;
    const els = [s('rect', { x: 60, y: 60, width: 300,
height: 110, rx: 8, class: 'ic-body' }), s('path', { d:
'M60 100 a15 15 0 0 1 0 30', class: 'ic-notch' }),
        s('text', { x: 210, y: 112, 'text-anchor':
'middle', class: 'ic-name' }, `IC ${ic.name}`),
s('text', { x: 210, y: 132, 'text-anchor': 'middle',
class: 'ic-title' }, ic.title)];
    for (let p = 1; p ⩽ 14; p++) {
        const bottom = p ⩽ 7;
        const x = bottom ? 80 + (p - 1) * 43.3 : 80 +
(14 - p) * 43.3;
        const y = bottom ? 170 : 40;
        const isUsed = used[p] ≠≠ undefined || p ⩵⩵ 7
|| p ⩵⩵ 14;
        const role = p ⩵⩵ 7 ? 'GND' : p ⩵⩵ 14 ? '+5 V'
: used[p];
        els.push(
            s('rect', { x: x - 9, y, width: 18, height:
20, rx: 2, class: `ic-pin${isUsed ? ' is-used' : ''}${p
⩵⩵ 14 ? ' is-vcc' : ''}${p ⩵⩵ 7 ? ' is-gnd' : ''}` }),
            s('text', { x, y: bottom ? 158 : 76, 'text-
anchor': 'middle', class: 'ic-pin-label' }, ic.pins[p]),
            s('text', { x, y: bottom ? 205 : 33, 'text-
anchor': 'middle', class: 'ic-pin-num' }, String(p)));
        if (role) els.push(s('text', { x, y: bottom ?
222 : 16, 'text-anchor': 'middle', class: 'ic-role' },
role));
    }
    return s('svg', { viewBox: `0 0 ${w} 232`, class:
'ic-svg', role: 'img', 'aria-label': `IC ${ic.name}
pinout` }, ...els);
}
```

One function draws any of the six TTL chips. Pins 1–7 run left to right along the bottom and 8–14 right to left along the top, as on a real DIP package seen from above with the notch on the left. Pins passed in `used` are highlighted and labelled with what they connect to; VCC and GND are always marked.

What was checked

Three kinds of test were run: the models were checked against hand calculations and truth tables in Node.js, every page and tab was opened in a real browser to look for errors, and every experiment was carried out end to end through its buttons.

1. Model tests (Node.js)

WHAT	CHECK	RESULT
Two-source DC network	KCL at node A and KVL around the left loop, two sets of values	**Pass**
Experiment 2 T-network	Vs − VR1 − VR2 = 0 and I = I2 + I3 for Vs = 10 V	**Pass**
Series RLC at resonance	Z = R and power factor = 1 at f0 = 1/(2π√LC)	**Pass**
Op-amp	Inverting gain −10 gives −5 V for 0.5 V in; gain −100 saturates at −13.5 V	**Pass**
RC filters	Low-pass and high-pass gain = 0.707 at the cut-off frequency	**Pass**
Diode bench	Vd + I·R = Vs, and the forward drop at 1.7 V supply lies between 0.6 and 0.8 V	**Pass**
Full adder, full subtractor, 4:1 MUX	Every row of the simulated truth table compared with arithmetic (S + 2·Cout = A + B + Cin, etc.)	**Pass**
Six universal-gate circuits (Exp 4)	Every row compared with NOT, AND, OR	**Pass**
K-map minimisation	m(1, 3, 7, 11, 15) minimises to A'B'D + CD	**Pass**
JK flip-flop (Exp 6)	All 8 combinations of Q, J, K with a two-delay pulse give the expected next state with Q' = not Q	**Pass**
Race-around	J = K = 1 with the clock held high never settles	**Pass**

124 individual checks, 124 passed.

2. Every page in the browser

An automated browser (Chromium via Playwright) opened all 22 pages and clicked through all 87 tabs, recording any JavaScript error or console error. Result: no errors. The screenshots were reviewed by eye, which found and fixed several problems: circuit labels overlapping symbols, a schematic cut off at the top when a gate sat above the first input, labels colliding at the end of the ABS graph, the diode's reverse current plotted on the same scale as its forward current, and the sticky top bar covering screenshots.

3. Experiments end to end

EXPERIMENT	DONE THROUGH THE BENCH	RESULT WRITTEN BY THE PAGE
Exp 3	Full set of forward and reverse readings	Knee ≈ 0.60 V, static and dynamic resistance calculated
Exp 4	All six circuits, every input combination	All six truth tables verified
Exp 5	Half and full adder, every combination	Both truth tables verified
Exp 6	Set, hold, reset, toggle, toggle; then clock held high	Truth table verified; race-around recorded with the number of transitions

EXPERIMENT	DONE THROUGH THE BENCH	RESULT WRITTEN BY THE PAGE
Exp 7	Two readings per project, one each side of the threshold	Output switched correctly at the threshold for all three projects

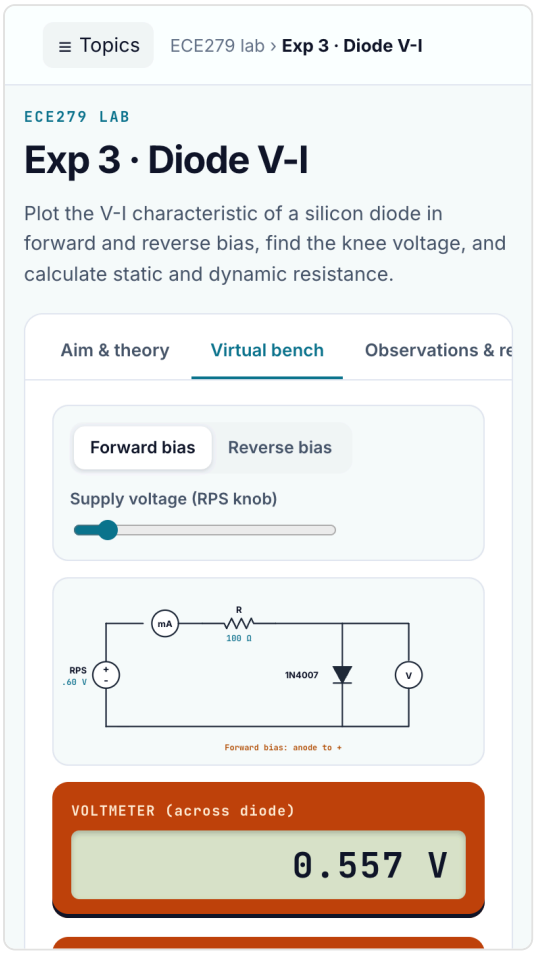


Fig. 76 · **Phone width.** The sidebar collapses behind a menu button and the bench stacks vertically.

Online and what I learned

Deployment

The code is in a public GitHub repository and served by GitHub Pages straight from the main branch. Because there is no build step, the files in the repository are exactly the files the browser loads.

Live site	https://lushadowx.github.io/ElectroBench/
Source code	https://github.com/LuShadowX/ElectroBench
Run locally	`python3 -m http.server 8000` in the project folder, then open `http://localhost:8000`

What I learned

- **Writing the model forces you to understand the topic.** To make the diode bench give sensible readings I had to solve the diode and resistor together, which is exactly the load-line idea from Unit 1. To make the flip-flop toggle once per pulse I had to think about propagation delay, which is what race-around is really about.
- **Real readings are not perfect.** Adding resistor tolerance and meter noise to Experiment 2 made the KVL sums come out at a few millivolts instead of zero, which is what happens on the real bench and is worth discussing in the viva.
- **Pin numbers matter.** Drawing the 7402 made it obvious why circuits copied from a 7400 diagram fail: the NOR chip has its outputs on different pins.
- **The two syllabus versions overlap a lot.** Mapping ECE249 and ECE131 side by side showed that the newer course trades digital electronics for circuits, machines and op-amps, so both were built into one site.

Possible extensions

- A drag-and-drop breadboard where the student places the chips and wires and the simulator checks the circuit.
- Connecting a real Arduino over Web Serial so Experiment 7 can plot live sensor values.
- Exporting the lab record as a filled lab-file template.